

Graduate Quantum Mechanics

Minimum Review and Exam Equations

Course lecture extracts: Chapters 1–3.9

Purpose. A compact, section-mapped review for a reader who has studied graduate quantum mechanics before. Retain the equations needed to reconstruct results, the assumptions under which they apply, and the distinctions most likely to cause mistakes. Exam-useful means useful for solving problems in this material; it is not a prediction of the instructor’s examination.

Scope. Based on 74 supplied physical lecture pages: `extract1.pdf` pp. 1–57 plus the continuation pp. 58–74. They contain Chapter 1 (1.1–1.12), Chapter 2 (2.1–2.13), and all of Chapter 3 (3.1–3.9). Source-page labels below refer to the lecture’s physical page numbers.

How to use. First identify the problem’s Hilbert space, basis, observable or Hamiltonian, and assumptions. Then select the relevant formula, perform the calculation, and check normalization, units, limiting cases, and phases. Attempt each self-check before consulting its answer. Small historical subsections are grouped, with original section numbers retained.

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Chapter 1: The Birth of Quantum Mechanics—Fundamental Equations

Minimum review of all lecture subsections 1.1–1.12, including 1.5.1. Source equations are retained in the lecture’s conventions; paragraphs marked **Supplement** add reconstruction steps or safeguards, not extra lecture claims. This is a review guide, not a prediction of examination questions.

1.1, 1.11, 1.12 Historical frame and structural synthesis

Source PDF pp. 1, 12–13.

Keep the structure, not the biographies. With $\hbar = h/(2\pi)$, action becomes phase in units of $\hbar$. The old quantum theory supplied discrete spectra and quantized action but no general consistent dynamics. By the 1927 Solvay meeting, wave and matrix mechanics had consolidated into equivalent descriptions. Correct equations often preceded their interpretations (Planck quanta, Born probabilities, Dirac antiparticles); the later path integral added another formulation.

Structural step	What must be reconstructed
Energy/momentum and waves	$E = \hbar\omega$, $\mathbf{p} = \hbar\mathbf{k}$
Noncommutative kinematics	$[\hat{q}, \hat{p}] = i\hbar$
Dynamics versus measurement	Schrödinger evolution versus Born probabilities
Representation independence	Matrices and wave functions describe the same state
Relativity / classical limit	Spinors / stationary action in the path integral

Operators emphasize algebra and symmetry; wave functions emphasize boundary-value problems; paths emphasize spacetime and semiclassical reasoning. No single equation alone supplies the full theory.

1.2 Planck’s law and the appearance of $\hbar$

Source PDF pp. 2–3.

For equilibrium electromagnetic radiation, $u_\nu(T) d\nu$ is energy per volume in $[\nu, \nu + d\nu]$. With c the speed of light and $k_B = R/N_A$,

$$u_\nu(T) = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/(k_B T)} - 1}, \quad \bar{E} = \frac{h\nu}{e^{h\nu/(k_B T)} - 1}.$$

Reconstruct: count mode excitations $E_n = nh\nu$, $n = 0, 1, \dots$, weight by $e^{-nh\nu/(k_B T)}$, and multiply $\bar{E}$ by mode density $8\pi\nu^2/c^3$. For $x = h\nu/(k_B T)$,

$$x \ll 1: \quad u_\nu \simeq \frac{8\pi\nu^2}{c^3} k_B T \quad (\text{Rayleigh–Jeans}), \quad x \gg 1: \quad u_\nu \simeq \frac{8\pi h\nu^3}{c^3} e^{-x} \quad (\text{Wien}).$$

Supplement—derivation/pitfalls: $Z = \sum_{n \geq 0} e^{-nx} = (1 - e^{-x})^{-1}$, so $\bar{n} = -\partial_x \ln Z = (e^x - 1)^{-1}$. This is excitation energy, not zero-point energy. u_ν is per ordinary frequency, not per wavelength or angular frequency; changing variable requires a Jacobian. Extending Rayleigh–Jeans to all ν makes $\int_0^\infty u_\nu d\nu$ diverge; the exponential tail removes this.

1.3 Planck–Einstein and de Broglie relations

Source PDF pp. 3–4.

$$E = h\nu = \hbar\omega, \quad \mathbf{p} = \hbar\mathbf{k}, \quad p = h/\lambda, \quad (E/c, \mathbf{p}) = \hbar(\omega/c, \mathbf{k}).$$

The photoelectric relation is $K_{\max} = h\nu - W$, with work function W . The plane wave $\psi \propto e^{i(\mathbf{k}\cdot\mathbf{r} - \omega t)} = e^{i(\mathbf{p}\cdot\mathbf{r} - Et)/\hbar}$ connects energy and momentum to phase gradients. A single-valued orbital phase motivates the old semiclassical condition $\oint p_i dq_i = n_i h$ (one action cycle; not an implicit sum). **Supplement:** emission requires $\nu \geq W/h$; a negative computed $K_{\max}$ means no photoemission. Distinguish ν from $\omega = 2\pi\nu$.

1.4 The old quantum theory as a bridge

Source PDF pp. 4.

$$E_n = -\frac{\text{Ry}}{n^2}, \quad n = 1, 2, \dots, \quad \text{Ry} \simeq 13.6 \text{ eV}, \quad h\nu = E_{n_i} - E_{n_f} = \text{Ry} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right).$$

These are nonrelativistic hydrogen energies; emission has $n_i > n_f$ and positive photon energy. The successful spectrum did not validate classical electron orbits as a general dynamics. **Supplement:** $\oint p_i dq_i = n_i h$ is the lecture's old quantization rule, not a universal exact condition (modern semiclassical rules can require turning-point phase corrections).

1.5 Matrix mechanics: noncommuting observables

Source PDF pp. 5–6.

Position and momentum are operators, not simultaneously assigned classical numbers. For canonical variables and a Hamiltonian $\hat{H}$,

$$\boxed{[\hat{q}, \hat{p}] = i\hbar \hat{I}}, \quad \boxed{\frac{d\hat{A}}{dt} = \frac{i}{\hbar} [\hat{H}, \hat{A}] + \left(\frac{\partial \hat{A}}{\partial t} \right)_{\text{explicit}}}.$$

The latter is the Heisenberg-picture equation; in particular, $\dot{\hat{q}} = (i/\hbar)[\hat{H}, \hat{q}]$ and $\dot{\hat{p}} = (i/\hbar)[\hat{H}, \hat{p}]$ when neither has explicit time dependence. **Sign check:** $(i/\hbar)[\hat{H}, \hat{A}] = [\hat{A}, \hat{H}]/(i\hbar)$; reversing the commutator without changing the prefactor is wrong.

Supplement—reconstruct Hamilton-like motion. Use $[AB, C] = A[B, C] + [A, C]B$. For $\hat{H} = \hat{p}^2/(2m) + V(\hat{q})$,

$$[\hat{p}^2, \hat{q}] = -2i\hbar \hat{p}, \quad [V(\hat{q}), \hat{p}] = i\hbar V'(\hat{q}) \implies \dot{\hat{q}} = \hat{p}/m, \quad \dot{\hat{p}} = -V'(\hat{q}).$$

These manipulations assume a common suitable operator domain and sufficient regularity of V . Commuting with $\hat{H}$ implies conservation only if the explicit time derivative also vanishes.

1.5.1 Why the commutator precedes uncertainty

For Hermitian observables in a normalized state with finite variances,

$$(\Delta A)^2 = \langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2, \quad \boxed{\Delta A \Delta B \geq \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|} \implies \Delta q \Delta p \geq \hbar/2.$$

The modulus is essential: the commutator expectation is generally imaginary. This concerns the spread of measurement outcomes in a state, not experimental clumsiness. **Supplement—proof backbone:** center the operators, $\delta A = \hat{A} - \langle \hat{A} \rangle$, $\delta B = \hat{B} - \langle \hat{B} \rangle$. Cauchy–Schwarz gives $\Delta A \Delta B \geq |\langle \delta A \delta B \rangle| \geq |\text{Im} \langle \delta A \delta B \rangle| = \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|$. A zero commutator expectation makes this lower bound zero; it need not make both variances zero or make the operators commute.

1.6 Wave mechanics: the Schrödinger equation

Source PDF pp. 7.

For one nonrelativistic particle in a scalar potential,

$$\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{r}), \quad \boxed{i\hbar \partial_t \psi(\mathbf{r}, t) = \hat{H} \psi(\mathbf{r}, t)}, \quad \hat{H} \psi(\mathbf{r}) = E \psi(\mathbf{r}).$$

Acting on the plane wave motivates $E \mapsto i\hbar \partial_t$ and $\mathbf{p} \mapsto -i\hbar \nabla$. Applying these to $E = p^2/(2m) + V$ motivates, but does *not derive*, the new dynamical postulate from classical mechanics. Spectra become differential-operator eigenvalue problems.

Supplement—assumptions/link: when $\hat{H}$ is time independent, write $\Psi_E(\mathbf{r}, t) = \psi_E(\mathbf{r}) e^{-iEt/\hbar}$. This converts the evolution equation into $\hat{H} \psi_E = E \psi_E$; a general state is a superposition, not one energy eigenstate. Self-adjoint $\hat{H}$ with appropriate boundary conditions generates norm-preserving unitary evolution. For time-independent $\hat{H}$, $U(t, t_0) = e^{-i\hat{H}(t-t_0)/\hbar}$. Do not use this simple exponential blindly for a time-dependent noncommuting Hamiltonian. The displayed $\hat{H}$ omits spin, relativistic effects, and magnetic vector potentials.

1.7 One quantum mechanics, many representations

Source PDF pp. 7–8.

In a complete orthonormal basis,

$$|\psi\rangle = \sum_n c_n |n\rangle, \quad c_n = \langle n | \psi \rangle, \quad A_{mn} = \langle m | \hat{A} | n \rangle, \quad \psi(\mathbf{r}) = \langle \mathbf{r} | \psi \rangle.$$

In position space, $\hat{\mathbf{p}} = -i\hbar\nabla$ yields $[\hat{r}_i, \hat{p}_j] = i\hbar\delta_{ij}\hat{I}$ by the product rule. The lecture uses the simultaneous unitary map

$$|\psi\rangle \mapsto U|\psi\rangle, \quad \hat{A} \mapsto U\hat{A}U^\dagger, \quad U^\dagger U = UU^\dagger = \hat{I}.$$

Supplement—invariance check: $\langle\psi|U^\dagger(U\hat{A}U^\dagger)U|\psi\rangle = \langle\psi|\hat{A}|\psi\rangle$; eigenvalues are unchanged as well. Do not transform the state alone and then claim invariance for a fixed observable. Passive basis conventions may put daggers in the opposite positions: be consistent.

1.8 Born rule, measurement, and Hilbert space

Source PDF pp. 9–10.

Evolution is not the measurement rule. For a normalized wave function, $\rho(\mathbf{r}, t) = |\psi(\mathbf{r}, t)|^2$ is a probability density, not a probability at a point. In an orthonormal eigenbasis of an observable,

$$\hat{A}|a_n\rangle = a_n|a_n\rangle, \quad |\psi\rangle = \sum_n c_n|a_n\rangle, \quad \boxed{P(a_n) = |c_n|^2 = |\langle a_n|\psi\rangle|^2}.$$

The displayed probability is for a nondegenerate eigenvalue (or a specified resolved basis state). Amplitudes interfere; only their squared moduli give probabilities. For a statistical ensemble of normalized states,

$$\boxed{\hat{\rho} = \sum_i p_i |\phi_i\rangle\langle\phi_i|}, \quad p_i \geq 0, \quad \sum_i p_i = 1, \quad \boxed{\langle A \rangle = \text{Tr}(\hat{\rho}\hat{A})}, \quad S = -k_B \text{Tr}(\hat{\rho} \ln \hat{\rho}).$$

For a pure state, $\hat{\rho} = |\psi\rangle\langle\psi|$ and $S = 0$. The ensemble vectors need not be mutually orthogonal.

Supplement—normalization and projectors. $P(\mathbf{r} \in \Omega) = \int_\Omega |\psi|^2 d^3r$ and $\int |\psi|^2 d^3r = 1$. For the full eigenspace of a possibly degenerate eigenvalue a , let P_a be its orthogonal projector. Ideal selective projective measurement has

$$\text{Pr}(a) = \text{Tr}(\hat{\rho}P_a), \quad \hat{\rho} \mapsto \frac{P_a \hat{\rho} P_a}{\text{Tr}(\hat{\rho}P_a)} \quad \text{if } \text{Pr}(a) > 0.$$

Thus degeneracy requires summing over the eigenspace, not choosing one arbitrary basis vector. An outcome with zero probability has no defined conditioned state. The projection rule is a measurement update, not ordinary closed-system Schrödinger evolution.

Supplement—pure versus mixed. Density operators are positive, Hermitian, and have trace one. If their eigenvalues are r_j , then $S = -k_B \sum_j r_j \ln r_j$, with $0 \ln 0 = 0$; $\text{Tr} \hat{\rho}^2 = 1$ for a pure state and is less than one for a mixed state. In an orthonormal two-state basis, the coherent state $(|0\rangle + |1\rangle)/\sqrt{2}$ and the equal incoherent mixture have matrices

$$\hat{\rho}_{\text{pure}} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad S = 0; \quad \hat{\rho}_{\text{mix}} = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad S = k_B \ln 2.$$

They have identical probabilities in this basis but different coherence and predictions in other bases. Ensemble weights p_i are not generally eigenvalues of $\hat{\rho}$ when the $|\phi_i\rangle$ are nonorthogonal.

Foundational limit (lecture): the projection postulate is operationally useful without deciding whether collapse is a literal physical process. Von Neumann's early hidden-variable argument used restrictive assumptions; Bell inequalities constrain local hidden-variable theories, not every conceivable additional-variable theory.

1.9 The Dirac equation: quantum mechanics meets relativity

Source PDF pp. 10–11.

Relativistic first-order time evolution requires a multicomponent spinor:

$$\boxed{i\hbar\partial_t\psi = (c\boldsymbol{\alpha} \cdot \hat{\mathbf{p}} + \beta mc^2)\psi}, \quad (i\hbar\gamma^\mu\partial_\mu - mc)\psi = 0, \quad \{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}\hat{I}.$$

The Clifford algebra ensures $E^2 = p^2c^2 + m^2c^4$. The lecture's key consequences are spinor structure, electron spin, hydrogen fine structure, and a negative-energy branch whose interpretation led to antiparticles. Relativistic structure is not obtained simply by changing E in the nonrelativistic Schrödinger equation.

Supplement—explicit convention and reconstruction. The lecture leaves the metric implicit; choose $g = \text{diag}(1, -1, -1, -1)$, $x^0 = ct$, $\beta = \gamma^0$, $\alpha^i = \gamma^0\gamma^i$. Then $\{\alpha^i, \alpha^j\} = 2\delta_{ij}\hat{I}$, $\{\alpha^i, \beta\} = 0$, and $\beta^2 = \hat{I}$ give, for the free Hamiltonian,

$$\hat{H}_D^2 = c^2\hat{\mathbf{p}}^2 + m^2c^4, \quad E = \pm\sqrt{p^2c^2 + m^2c^4}.$$

Cross terms vanish by anticommutation, not by setting the matrices to zero. Here $\partial_0 = c^{-1}\partial_t$, explaining the mc in the covariant equation versus mc^2 in the Hamiltonian. The simple squared free Hamiltonian should not be applied unchanged with interactions.

1.10 The Dirac–Feynman path integral: a third formulation

Source PDF pp. 11.

Core meaning. The propagator is the position-space matrix element of the same unitary evolution operator used in the Schrödinger formulation:

$$K(x_b, t_b; x_a, t_a) = \langle x_b | e^{-i\hat{H}(t_b-t_a)/\hbar} | x_a \rangle, \quad \psi(x_b, t_b) = \int dx_a K(x_b, t_b; x_a, t_a) \psi(x_a, t_a).$$

It is a transition *amplitude density*; one must add amplitudes over indistinguishable alternatives before taking an absolute square. For fixed endpoints, the lecture rewrites this same object schematically as

$$K(x_b, t_b; x_a, t_a) = \int_{x(t_a)=x_a}^{x(t_b)=x_b} \mathcal{D}x(t) e^{iS[x]/\hbar}, \quad S[x] = \int_{t_a}^{t_b} L(x, \dot{x}, t) dt.$$

Here a “path” is a complete intermediate history $x(t)$ satisfying the two endpoint conditions. The symbol $\mathcal{D}x$ is not an ordinary one-variable integral: it is defined by time slicing. Every path contributes a unit-modulus phase, not a classical probability; different paths interfere.

Supplement—derive the path sum from the operator propagator. Take $T = t_b - t_a = N\epsilon$, insert $N - 1$ position resolutions of identity, and use the composition law:

$$K = \lim_{N \rightarrow \infty} \int \prod_{j=1}^{N-1} dx_j \prod_{j=0}^{N-1} \langle x_{j+1} | e^{-i\epsilon \hat{H}/\hbar} | x_j \rangle, \\ x_0 = x_a, \quad x_N = x_b.$$

For $\hat{H} = \hat{p}^2/(2m) + V(\hat{x})$, first-order Trotter splitting and one momentum resolution give the short-time kernel

$$\langle x_{j+1} | e^{-i\epsilon \hat{H}/\hbar} | x_j \rangle = \int \frac{dp_j}{2\pi\hbar} e^{\frac{i}{\hbar}[p_j(x_{j+1}-x_j) - \epsilon(p_j^2/2m + V(x_j))]} + O(\epsilon^2) \\ = \left(\frac{m}{2\pi i \hbar \epsilon} \right)^{1/2} e^{\frac{i\epsilon}{\hbar} [\frac{m}{2}((x_{j+1}-x_j)/\epsilon)^2 - V(x_j)]} + O(\epsilon^2).$$

Consequently,

$$K = \lim_{N \rightarrow \infty} \left(\frac{m}{2\pi i \hbar \epsilon} \right)^{N/2} \int \prod_{j=1}^{N-1} dx_j e^{iS_N/\hbar}, \quad S_N = \sum_{j=0}^{N-1} \epsilon \left[\frac{m}{2} \left(\frac{x_{j+1} - x_j}{\epsilon} \right)^2 - V(x_j) \right].$$

Thus “sum over all paths” means an ordinary multiple integral over every intermediate position, followed by $N \rightarrow \infty$. The prefactor is part of $\mathcal{D}x$; omitting it loses normalization, dimensions, and the $T \rightarrow 0$ limit $K \rightarrow \delta(x_b - x_a)$. The displayed discretization is for the standard scalar-potential Hamiltonian; velocity-dependent interactions require care over operator ordering and the time-slicing prescription.

Why the classical path emerges without deleting quantum paths. Actions differing by much more than $\hbar$ typically produce rapidly oscillating relative phases and cancellation. For a variation $x \mapsto x + \eta$ with fixed endpoints, $\eta(t_a) = \eta(t_b) = 0$, integration by parts gives

$$\delta S = \int_{t_a}^{t_b} \left(\frac{\partial L}{\partial x} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}} \right) \eta dt.$$

Hence $\delta S = 0$ gives the Euler–Lagrange equation. Near such a path, $S[x_{cl} + \eta] = S[x_{cl}] + \frac{1}{2} \delta^2 S + \dots$: the linear phase variation vanishes, so nearby contributions remain coherent to leading order. Nonstationary neighborhoods cancel more strongly. The criterion concerns action *variations* relative to $\hbar$; adding a large constant to all actions does not create a classical limit. Stationary does not mean a unique path or an absolute minimum. The functional integral still includes every path; classical mechanics appears by stationary-phase interference, not because nonclassical paths acquire zero probability.

Anchor example—free particle. For $V = 0$ the Gaussian time-sliced integrals give exactly

$$K_0(x_b, t_b; x_a, t_a) = \sqrt{\frac{m}{2\pi i \hbar T}} \exp \left[\frac{i}{\hbar} \frac{m(x_b - x_a)^2}{2T} \right], \quad T > 0.$$

The exponent is the classical action of the straight path, while the prefactor is the fluctuation contribution from paths around it. Checks: $[K_0] = L^{-1}$, $K_0 \rightarrow \delta(x_b - x_a)$ distributionally as $T \rightarrow 0^+$, and convolution of two free kernels gives the kernel for the summed time interval.

Self-check: reconstruct before consulting the key

1. **1.1, 1.11–1.12:** Why are old orbit quantization rules insufficient? Name the distinct roles of dynamics, measurement, and representation.
2. **1.2:** Starting with $Z = (1 - e^{-x})^{-1}$, obtain the mean mode energy and both limits of u_ν . Which extrapolation causes the ultraviolet catastrophe?
3. **1.3–1.4:** Recover the energy and momentum operators from the plane wave. For hydrogen emission, which is positive: $E_{n_i} - E_{n_f}$ or its negative? What does $h\nu < W$ imply for photoemission?
4. **1.5:** For $\hat{H} = \hat{p}^2/(2m) + V(\hat{q})$, derive both Heisenberg canonical equations and state the extra condition for $[\hat{H}, \hat{A}] = 0$ to imply that $\hat{A}$ is conserved.
5. **1.5.1:** What does $\Delta q \Delta p \geq \hbar/2$ describe? Does $\langle [\hat{A}, \hat{B}] \rangle = 0$ imply $[\hat{A}, \hat{B}] = 0$?
6. **1.6:** When may you separate off $e^{-iEt/\hbar}$? Why is the operator-substitution argument not a classical derivation of quantum mechanics?
7. **1.7:** Show an expectation value is invariant under the lecture's simultaneous unitary map. Is a wave function a different physical theory from a matrix?
8. **1.8:** How do you compute and condition on a degenerate measurement outcome? Can a pure superposition and a mixture agree in one measurement basis?
9. **1.9:** Which anticommutation relations remove the cross terms in $\hat{H}_D^2$? Why does the covariant equation contain mc rather than mc^2 ?
10. **1.10:** Starting from $\langle x_b | e^{-i\hat{H}T/\hbar} | x_a \rangle$, where do the integrations over intermediate positions enter? Why are path *amplitudes*, rather than path probabilities, added? What endpoint condition is used in varying the action, and what must the kernel approach as $T \rightarrow 0^+$?

Compact answers

1. They mix classical trajectories with imposed quantum rules and lack general dynamics. Schrödinger gives evolution; Born gives probabilities; unitary representation changes preserve physical predictions.
2. $\bar{E} = -h\nu \partial_x \ln Z = h\nu/(e^x - 1)$. For small x , $e^x - 1 \simeq x$; for large x , $e^x - 1 \simeq e^x$. Extending the resulting $u_\nu \propto \nu^2 T$ to infinite frequency diverges.
3. $i\hbar \partial_t \psi = E\psi$ and $-i\hbar \nabla \psi = \mathbf{p}\psi$. Emission: $n_i > n_f$, $E_{n_i} - E_{n_f} > 0$. Below threshold: no photoemission.
4. $[\hat{p}^2, \hat{q}] = -2i\hbar \hat{p}$ and $[V, \hat{p}] = i\hbar V'$ give $\dot{\hat{q}} = \hat{p}/m$, $\dot{\hat{p}} = -V'$. Also require no explicit time dependence.
5. Intrinsic ensemble spreads in the specified state, not apparatus error. No: a particular expectation can vanish for a nonzero operator.
6. For an energy eigenstate of time-independent $\hat{H}$. Substitution motivates a new dynamical law; classical mechanics does not supply that postulate.
7. Cancel $U^\dagger U$ in the transformed expectation. No: wave functions and matrices are representations of the same abstract theory.
8. $\text{Pr}(a) = \text{Tr}(\hat{\rho} P_a)$ and $\hat{\rho}_a = P_a \hat{\rho} P_a / \text{Pr}(a)$. Yes; coherences distinguish them in other bases.
9. $\{\alpha^i, \alpha^j\} = 2\delta_{ij} \hat{I}$, $\{\alpha^i, \beta\} = 0$, $\beta^2 = \hat{I}$. With $x^0 = ct$, multiplying the covariant equation by $c\gamma^0$ gives Hamiltonian units.
10. Insert $N - 1$ identities $\int dx_j |x_j\rangle \langle x_j|$ between short-time evolution factors. Alternatives are indistinguishable, so complex amplitudes add before $|\cdot|^2$; their relative phases cause interference. Endpoint variations vanish, $\eta(t_a) = \eta(t_b) = 0$, and $K \rightarrow \delta(x_b - x_a)$ distributionally.

Chapter 2: Experimental Clues and Fundamental Concepts

Exam minimum, source PDF pp. 15–40. Keep $\hbar = h/(2\pi)$ explicit; SI units unless stated otherwise. $h, \hbar$ have units of action, spin has units of angular momentum, and Pauli matrices are dimensionless. Hats denote operators when distinction from their matrix representations matters. Discrete-spectrum statements below assume a complete orthonormal eigenbasis; differential operators require appropriate domains and boundary conditions.

2.1 From experimental facts to the quantum description

Source PDF pp. 15–16.

2.1.1 Experimental map. Interference/diffraction of light *and matter* demand amplitudes and phases; black-body radiation, photoemission and Compton scattering demand quanta; spectra and Franck–Hertz demand discrete internal energies; Stern–Gerlach (SG) demands discrete spin-component outcomes, preparation, and incompatible measurements. These are constraints on a theory, not a derivation of the state space or Born rule. Wave/particle behavior refers to experimental outcomes, not a switch between two classical objects.

2.2 Several experiments that reveal quantum behavior

Source PDF pp. 16–19.

2.2.1 Black-body radiation. An ideal absorber/emitter in thermal equilibrium has spectral radiance

$$B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/(k_B T)} - 1}, \quad E_n = n h \nu \quad (n = 0, 1, \dots).$$

Here B_ν is per unit frequency, with units $\text{W m}^{-2} \text{sr}^{-1} \text{Hz}^{-1}$; do not use this as a per-wavelength spectrum without a Jacobian. Quantized *exchange* in units $h\nu$ defeats classical equipartition at high frequency. This historical E_n convention is not the later harmonic oscillator's full $(n + \frac{1}{2})\hbar\omega$ spectrum. *Use:* form $u = h\nu/(k_B T)$; for $u \ll 1$, $B_\nu \simeq 2\nu^2 k_B T / c^2$, while $u \gg 1$ yields exponential suppression.

2.2.2 Photoelectric effect. With positive elementary charge e and work function Φ ,

$$E_\gamma = h\nu, \quad K_{\max} = h\nu - \Phi = eV_s, \quad \nu_0 = \Phi/h.$$

V_s denotes the positive *magnitude* of the stopping voltage. In the single-photon regime no emission occurs for $\nu < \nu_0$. Above threshold, frequency controls maximum electron energy; intensity primarily controls photon arrival rate and hence current, not $K_{\max}$. *Use:* the slope of V_s versus ν is h/e and the intercept is $-\Phi/e$. Use joules throughout, or convert consistently to electron-volts.

2.2.3 Compton scattering. For a photon scattering through angle θ from an initially stationary, effectively free electron,

$$E_\gamma = pc = h\nu, \quad p_\gamma = h/\lambda, \quad \boxed{\lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \theta)}.$$

Energy *and vector momentum* conservation, with relativistic electron energy, give this result. $\lambda_C = h/(m_e c) \simeq 2.426 \times 10^{-12} \text{ m}$. The shift is zero forward, λ_C at 90° , and $2\lambda_C$ backward; λ' is the outgoing photon wavelength. Do not substitute the electron recoil angle or assume the same formula for a tightly bound/moving electron.

2.2.4 Franck–Hertz. Accelerated electrons in a dilute vapor lose energy in inelastic atomic excitation only when they can supply an allowed spacing

$$\Delta E = E_n - E_m > 0, \quad \Delta E \simeq e \Delta V.$$

Successive current drops/recoveries indicate repeated excitations. Use the *spacing* of voltage features to infer level gaps; voltage offsets and collection geometry can move absolute thresholds. Low-energy predominantly elastic collisions do not produce these discrete losses.

2.2.5 What is established? Discrete exchanges, wave-like and particle-like phenomena, discrete atomic internal energies, and h as the wave/mechanical conversion scale. These experiments alone do not prescribe quantum states, observable operators, or outcome probabilities.

2.3 The Stern–Gerlach experiment

Source PDF pp. 19–24.

2.3.1 Two-state system and deflection. Neutral Ag has configuration $[\text{Kr}] 4d^{10}5s^1$: closed shells give no net electronic angular momentum, and the outer electron has $l = 0$, $s = 1/2$. In the course’s effective two-state approximation,

$$H_{\text{int}} = -\boldsymbol{\mu} \cdot \mathbf{B}, \quad \mathbf{F} = \nabla(\boldsymbol{\mu} \cdot \mathbf{B}), \quad F_z \simeq \mu_z \partial_z B_z.$$

A uniform field can produce torque but not this spatial splitting. For magnet length L , atom mass M , approximately constant longitudinal speed v , weak deflection and approximately constant gradient,

$$\Delta t \simeq L/v, \quad \Delta v_z \simeq F_z L/(Mv), \quad \theta_z \simeq \frac{\mu_z L \partial_z B_z}{Mv^2} = \frac{\mu_z L \partial_z B_z}{2E_k}.$$

Here $E_k = Mv^2/2$ is *translational*, not spin energy. A downstream drift distance D gives $z \simeq F_z L(D + L/2)/(Mv^2)$ if initial transverse velocity is zero. Thermal $E_k \sim 3k_B T/2$ is only a scale estimate: an effusive beam has a velocity distribution. Classically $\mu_z = \mu \cos \theta$ would span a continuum; experiment gives two branches identified with $S_z = \pm \hbar/2$ in the modern theory. Branch displacement measures μ_z ; an electron’s magnetic moment is opposite its spin, so do not label the upper spot “spin up” without the sign of the coupling and gradient. SG is not a precision determination of $\hbar$.

2.3.2 Sequential analyzers. Selecting the $z+$ branch prepares $|z+\rangle$. Then

$$P(z+|z+) = 1, \quad P(x\pm|z+) = \frac{1}{2}, \quad P(z\pm|x+) = \frac{1}{2}.$$

For $z+ \rightarrow x+ \rightarrow z-$ the final conditional probability given the selected $x+$ beam is $1/2$, but the joint transmission probability relative to the initial $z+$ beam is $1/4$. Multiply conditional probabilities for a sequence of *resolved/selected measurements*; add amplitudes for coherent alternatives that have not been distinguished. The intermediate x filter removes certainty of S_z .

2.3.3 Classical polarization analogy. $\mathbf{E}_{\text{physical}} = \text{Re}[(E_x \hat{\mathbf{x}} + E_y \hat{\mathbf{y}})e^{i(kz - \omega t)}]$. Identify $|z+\rangle \leftrightarrow \hat{\mathbf{x}}$ and $|z-\rangle \leftrightarrow \hat{\mathbf{y}}$. Equal real superpositions represent $\pm 45^\circ$ linear polarization; $(1, \pm i)/\sqrt{2}$ represent circular polarization. Circular handedness labels depend on viewing/time conventions. This is a classical two-component complex-vector analogy, not an appeal to photons. The source’s $|x-\rangle = (-|z+\rangle + |z-\rangle)/\sqrt{2}$ here differs by overall minus sign from its later convention; both represent the same ray.

2.4 State vectors and Dirac notation

Source PDF pp. 24–26.

2.4.1 Kets, bras, inner products. A pure state is a ray: $|\alpha\rangle \sim c|\alpha\rangle$, $c \neq 0$; choose $\langle\alpha|\alpha\rangle = 1$. Only a global phase remains between normalized representatives. Conjugate *all* coefficients when forming a bra:

$$|\psi\rangle = \sum_n c_n |n\rangle \Rightarrow \langle\psi| = \sum_n c_n^* \langle n|, \quad \langle\beta|\alpha\rangle = \langle\alpha|\beta\rangle^*, \quad \langle\alpha|\alpha\rangle \geq 0.$$

Zero inner product means orthogonality; zero norm means the zero vector, not a physical state. Inner products are linear in the ket, conjugate-linear in the bra.

2.4.2 Operators and adjoints. $A(c|\alpha\rangle + d|\beta\rangle) = cA|\alpha\rangle + dA|\beta\rangle$; $\langle\beta|A|\alpha\rangle = \langle\alpha|A^\dagger|\beta\rangle^*$; $(AB)^\dagger = B^\dagger A^\dagger$. Observables are Hermitian ($A^\dagger = A$; self-adjointness matters for unbounded operators). An outer product is an operator: $(|\alpha\rangle\langle\beta|)|\gamma\rangle = |\alpha\rangle\langle\beta|\gamma\rangle$, unlike the scalar inner product.

2.5 Observables, eigenstates, and matrix representations

Source PDF pp. 26–28.

2.5.1 Hermitian eigenproblem. $A|a_n\rangle = a_n|a_n\rangle$. Taking the expectation and its conjugate proves $a_n = a_n^*$. For two eigenvectors,

$$(a_n - a_m)\langle a_m|a_n\rangle = 0.$$

Distinct eigenvalues force orthogonality; equal eigenvalues do not. Choose an orthonormal basis inside each degenerate eigenspace. With r indexing that basis,

$$\begin{aligned} I &= \sum_{a,r} |a,r\rangle\langle a,r|, & P_a &= \sum_r |a,r\rangle\langle a,r|, \\ A &= \sum_a aP_a, & P_a P_b &= \delta_{ab} P_a, \quad P_a^\dagger = P_a. \end{aligned}$$

For nondegenerate levels, $P_n = |a_n\rangle\langle a_n|$. Inserting completeness is the standard way to derive expansions, spectral decompositions, and products:

$$|\psi\rangle = \sum_n |n\rangle\langle n|\psi\rangle = \sum_n c_n |n\rangle, \quad c_n = \langle n|\psi\rangle, \quad \sum_n |c_n|^2 = 1.$$

Pitfall: “distinct eigenvectors are orthogonal” is false without distinct eigenvalues or an explicit orthonormal choice.

2.5.2 Matrix representation. Fix the basis order before doing any algebra. State columns contain c_n , bras are conjugate-transposed rows, and

$$A_{mn} = \langle m|A|n\rangle, \quad (A\psi)_m = \sum_n A_{mn}c_n, \quad (AB)_{mn} = \sum_k A_{mk}B_{kn}, \quad (A^\dagger)_{mn} = A_{nm}^*.$$

Column n is the expansion of $A|n\rangle$. Reconstruct an operator from a matrix by $A = \sum_{mn} A_{mn}|m\rangle\langle n|$. Matrix mechanics is the same operator theory in coordinates, not another physical theory.

2.6 Measurement and the Born rule

Source PDF pp. 28–29.

2.6.1 Ideal projective measurement. For a normalized premeasurement pure state,

$$p(a) = \langle \psi|P_a|\psi\rangle = \|P_a|\psi\rangle\|^2, \quad |\psi_a\rangle = \frac{P_a|\psi\rangle}{\sqrt{p(a)}} \quad [p(a) > 0].$$

For a nondegenerate level, $p(a_n) = |\langle a_n|\psi\rangle|^2$ and the final ray is $|a_n\rangle$. For degeneracy, project onto the *whole* eigenspace; a measurement of A alone need not select a particular basis vector within it. This is the ideal Lüders update, not a claim about every real measuring device. A zero-probability outcome has no defined conditional state.

$$\sum_a p(a) = 1, \quad \langle A \rangle = \langle \psi|A|\psi\rangle = \sum_a a p(a), \quad \langle A^2 \rangle = \sum_a a^2 p(a).$$

For an unnormalized input, divide expectation values and probabilities by $\langle \psi|\psi\rangle$. An expectation is an ensemble average and need not be an allowed single-shot eigenvalue.

2.6.2 Selective preparation and phase. A selected SG output prepares an eigenstate. If $|\psi\rangle = c_+|+\rangle + c_-|-\rangle$, $c_\pm$ are amplitudes, not probabilities. Measuring S_z sees only their magnitudes; other bases reveal their relative phase. For a subsequent projector Q_b ,

$$p(a \text{ then } b) = \|Q_b P_a |\psi\rangle\|^2, \quad p(b|a) = \frac{\|Q_b P_a |\psi\rangle\|^2}{p(a)}.$$

Recipe: normalize input $\rightarrow$ identify the actual measured eigenspaces $\rightarrow$ project $\rightarrow$ compute norm squared $\rightarrow$ renormalize before the next conditional prediction. A selective measurement is not a passive relabeling of coordinates.

2.7 Spin-1/2 and the Pauli matrices

Source PDF pp. 29–31.

2.7.1 S_z basis and ladders. Throughout use the ordered basis $(|+\rangle, |-\rangle) = (|z+\rangle, |z-\rangle)$:

$$|+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |-\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad S_z = \frac{\hbar}{2}(|+\rangle\langle+| - |-\rangle\langle-|), \quad S_+ = \hbar|+\rangle\langle-|, \quad S_- = \hbar|-\rangle\langle+|.$$

$S_+|-\rangle = \hbar|+\rangle$, $S_-|+\rangle = \hbar|-\rangle$; both annihilate the other basis ket. Ladder operators are not Hermitian observables; $S_+^\dagger = S_-$.

2.7.2 Other axes and conventions.

$$|x\pm\rangle = \frac{|+\rangle \pm |-\rangle}{\sqrt{2}}, \quad |y\pm\rangle = \frac{|+\rangle \pm i|-\rangle}{\sqrt{2}}, \quad S_i = \frac{\hbar}{2}\sigma_i,$$

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The eigenvalues of σ_i are ± 1 , whereas those of S_i are $\pm\hbar/2$. Equal z -probabilities alone do not fix the relative phase defining an x axis; orthogonality plus a chosen axis/phase convention gives these expressions.

2.7.3 Spin algebra. Summation over repeated k is understood, with $\epsilon_{xyz} = +1$ and antisymmetry:

$$\begin{aligned} \sigma_i \sigma_j &= \delta_{ij} I + i\epsilon_{ijk} \sigma_k, & [S_i, S_j] &= i\hbar \epsilon_{ijk} S_k, \\ \{S_i, S_j\} &= \frac{\hbar^2}{2} \delta_{ij} I, & S_\pm &= S_x \pm iS_y, \\ [S_z, S_\pm] &= \pm \hbar S_\pm, & [S_-, S_+] &= -2\hbar S_z, \\ S_i^2 &= \frac{\hbar^2}{4} I, & S^2 &= \frac{3}{4} \hbar^2 I = s(s+1)\hbar^2 I \quad (s = \tfrac{1}{2}). \end{aligned}$$

Useful reduction: $(\mathbf{a} \cdot \boldsymbol{\sigma})(\mathbf{b} \cdot \boldsymbol{\sigma}) = (\mathbf{a} \cdot \mathbf{b})I + i(\mathbf{a} \times \mathbf{b}) \cdot \boldsymbol{\sigma}$. “Fermion/boson” concerns particle statistics, not whether an operator has a commutator or anticommutator.

Fast spin probabilities. For normalized $(c_+, c_-)^T$,

$$\begin{aligned} P(x\pm) &= \tfrac{1}{2}|c_+ \pm c_-|^2, & P(y\pm) &= \tfrac{1}{2}|c_+ \mp ic_-|^2, \\ \langle S_x \rangle &= \hbar \operatorname{Re}(c_+^* c_-), & \langle S_y \rangle &= \hbar \operatorname{Im}(c_+^* c_-), \\ \langle S_z \rangle &= \tfrac{\hbar}{2}(|c_+|^2 - |c_-|^2), & \langle \Delta S_i \rangle^2 &= \hbar^2/4 - \langle S_i \rangle^2. \end{aligned}$$

The opposite sign in the y amplitude comes from conjugating the eigenket.

2.8 Compatible and incompatible observables

Source PDF pp. 31.

In the discrete finite-dimensional setting, Hermitian A, B admit a complete common orthonormal eigenbasis iff $[A, B] = 0$. In an A eigenbasis,

$$(a_m - a_n)\langle a_m | B | a_n \rangle = 0.$$

If A is nondegenerate this forces B diagonal. If A is degenerate it forces B to be block diagonal; diagonalize B *within each equal- a block*. A common eigenvector need not imply commuting operators; noncommuting operators may have isolated common eigenvectors (e.g. all L_i annihilate an $l=0$ state). Spin-1/2 S_x, S_z have no common eigenvector and $[S_x, S_z] = -i\hbar S_y$. For unbounded operators, formal commutators alone require domain/spectral care.

2.9 Change of basis and unitary transformations

Source PDF pp. 32–33.

2.9.1 Same physical state, new coordinates. In the conventions of 2.7,

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad c_x = U^\dagger c_z, \quad A_x = U^\dagger A_z U.$$

The columns of U are $|x+\rangle, |x-\rangle$ expressed in the old z basis. Here $U = U^\dagger$, accidentally making forward and inverse look identical. Explicitly,

$$S_x^{(x)} = \frac{\hbar}{2}\sigma_z, \quad S_y^{(x)} = -\frac{\hbar}{2}\sigma_y, \quad S_z^{(x)} = \frac{\hbar}{2}\sigma_x.$$

The minus sign for $S_y^{(x)}$ is required by the chosen phases. Rephasing a basis vector changes matrix entries but not observables or probabilities.

2.9.2 General unitary basis map. For complete orthonormal bases $\{|a_n\rangle\}$ and $\{|b_n\rangle\}$,

$$\hat{U} = \sum_n |b_n\rangle\langle a_n|, \quad \hat{U}|a_n\rangle = |b_n\rangle, \quad U_{mn} = \langle a_m|b_n\rangle, \quad U^\dagger U = U U^\dagger = I.$$

The matrix U contains new basis vectors in old coordinates. Insert completeness to reconstruct the transformation rather than memorizing its direction:

$$(c_b)_n = \langle b_n|\psi\rangle = \sum_m \langle b_n|a_m\rangle (c_a)_m \Rightarrow \boxed{c_b = U^\dagger c_a, \quad A_b = U^\dagger A_a U}.$$

Unitary transformations preserve inner products, norms and probabilities. A *passive* basis change leaves the physical ket fixed and transforms its coordinates by $U^\dagger$. An *active* transformation sends $|\psi\rangle \rightarrow \hat{U}|\psi\rangle$ at fixed basis, hence $c \rightarrow Uc$. Do not mix these operations; similarly $UAU^\dagger$ is the actively conjugated operator, not the passive formula with the above definition of U .

2.9.3 Trace and diagonalization.

$$\text{Tr } A = \sum_n \langle a_n|A|a_n\rangle, \quad \text{Tr}(AB) = \text{Tr}(BA), \quad \text{Tr}(U^\dagger AU) = \text{Tr } A.$$

Cyclicity permits cyclic permutations, not arbitrary reordering. For example $\text{Tr}(ABC) = \text{Tr}(BCA)$, but not generally $\text{Tr}(ACB)$. These statements require finite matrices or suitable trace-class products; do not apply a finite trace argument to $[x, p] = i\hbar I$ on an infinite-dimensional space.

To diagonalize Hermitian B :

1. Solve $\det(B - bI) = 0$ for eigenvalues.
2. Solve $BC^{(l)} = b_l C^{(l)}$ for normalized eigenvectors; orthonormalize inside degenerate subspaces.
3. Put these columns into U : $U_{nl} = C_n^{(l)} = \langle a_n|b_l\rangle$.
4. Check $U^\dagger U = I$ and $U^\dagger BU = \text{diag}(b_l)$; use the same column order for eigenvalues, states and amplitudes.

Similarity preserves eigenvalues, trace and determinant. A basis change cannot alter predicted outcomes.

Mini-check of the recipe. The $z+$ state has old column $(1, 0)^T$ and new column $(1, 1)^T/\sqrt{2}$ in the x basis. It still has definite S_z : applying $S_z^{(x)} = (\hbar/2)\sigma_x$ to that column returns $(\hbar/2)$ times the same column. Its nontrivial coordinates in the x basis do not mean a measurement has occurred.

2.10 Continuous spectra: position and momentum

Source PDF pp. 33–36.

2.10.1 Generalized eigenstates and delta functions.

$$\hat{\xi}|\xi\rangle = \xi|\xi\rangle, \quad \langle\xi|\xi'\rangle = \delta(\xi - \xi'), \quad \int d\xi |\xi\rangle\langle\xi| = I, \quad |\psi\rangle = \int d\xi |\xi\rangle\langle\xi|\psi\rangle.$$

These are distributional eigenstates, not unit-norm Hilbert-space vectors. Integrals below run over the real line unless bounds are given. For smooth test functions,

$$\begin{aligned} \int dx \delta(x-a)f(x) &= f(a), & \int dx \delta'(x-a)f(x) &= -f'(a), \\ \delta(-x) &= \delta(x), & \delta(ax) &= \delta(x)/|a| \quad (a \neq 0), \\ x\delta(x) &= 0, & x\delta'(x) &= -\delta(x), \\ \delta(x^2 - a^2) &= \frac{\delta(x-a) + \delta(x+a)}{2|a|} \quad (a \neq 0), & \delta(x) &= \frac{1}{2\pi} \int dk e^{ikx}. \end{aligned}$$

A useful extension for simple roots x_j is $\delta(g(x)) = \sum_j \delta(x - x_j)/|g'(x_j)|$; it fails at multiple roots without further distributional analysis. $\delta(x)$ has units of inverse length. Do not treat $\delta(0)$ as a finite normalization or square a delta to obtain a probability.

2.10.2 Position representation.

$$\psi(x) = \langle x|\psi\rangle, \quad \int dx |\psi(x)|^2 = 1, \quad P(a < x < b) = \int_a^b dx |\psi(x)|^2, \quad (\hat{x}\psi)(x) = x\psi(x).$$

ψ has units $\text{length}^{-1/2}$ in one dimension. $|\psi(x)|^2$ is a density, not a probability at an exact point. A finite-resolution ideal position measurement uses an interval projector $P_{[a,b]} = \int_a^b dx |x\rangle\langle x|$.

2.10.3 Translation and its generator. Define an *active* translation by $T(a)|x\rangle = |x+a\rangle$. Then $T(0) = I$, $T(a)T(b) = T(a+b)$ and $T(a)^\dagger = T(-a) = T(a)^{-1}$.

$$T(\delta x) = I - iK\delta x + O(\delta x^2), \quad K^\dagger = K, \quad [x, K] = i, \quad p = \hbar K = -i\hbar\partial_x,$$

$$\boxed{T(a) = e^{-iap/\hbar}, \quad (T(a)\psi)(x) = \psi(x-a), \quad [x, p] = i\hbar I.}$$

The minus sign in the translated wave function shifts its peak to the *right* by a . Check $T^\dagger x T = x + aI$, or directly $[x, -i\hbar\partial_x]\psi = i\hbar\psi$. K has units inverse length; p has momentum units. Unitarity/strong continuity underlie the Hermitian generator; differentiability and operator domains are implicit.

2.10.4 Fourier-related wave functions. Choose the source's phase convention:

$$\begin{aligned} \phi(p) &= \langle p|\psi\rangle, & \langle x|p\rangle &= \frac{e^{ipx/\hbar}}{\sqrt{2\pi\hbar}}, \\ \psi(x) &= \frac{1}{\sqrt{2\pi\hbar}} \int dp e^{ipx/\hbar} \phi(p), & \phi(p) &= \frac{1}{\sqrt{2\pi\hbar}} \int dx e^{-ipx/\hbar} \psi(x). \end{aligned}$$

Here $\langle p|p'\rangle = \delta(p - p')$, $\int dp |\phi(p)|^2 = 1$, and ϕ has units $\text{momentum}^{-1/2}$. In momentum space p multiplies and $x = i\hbar\partial_p$. Use complex-conjugate Fourier kernels for inverse directions; changing from p to wave number $k = p/\hbar$ changes the normalization and measure. Integration by parts requires vanishing boundary terms on the chosen domain.

2.11 The uncertainty relation

Source PDF pp. 36–38.

2.11.1 Variance and Robertson inequality. Let $\delta A = A - \langle A \rangle I$ and $\Delta A = \sqrt{\langle (\delta A)^2 \rangle}$; the former is an operator, the latter a nonnegative number. For normalized states with the required finite moments/products,

$$\begin{aligned} (\Delta A)^2 &= \langle A^2 \rangle - \langle A \rangle^2, \\ (\Delta A)^2 (\Delta B)^2 &\geq |\langle \delta A \delta B \rangle|^2 = \frac{1}{4} |\langle [A, B] \rangle|^2 + \frac{1}{4} |\langle \{\delta A, \delta B\} \rangle|^2, \\ \boxed{\Delta A \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|}, \quad &\boxed{\Delta x \Delta p \geq \hbar/2}. \end{aligned}$$

Derivation: Cauchy–Schwarz on $\delta A|\psi\rangle, \delta B|\psi\rangle$; split their product into commutator plus anticommutator. Their expectations are respectively imaginary and real; drop the nonnegative covariance term for Robertson. Saturating Cauchy–Schwarz alone need not saturate Robertson: also require zero symmetrized covariance. For nonzero variances, Robertson equality requires the two fluctuation vectors proportional with a purely imaginary coefficient. A nonzero commutator can have zero expectation in a particular state, giving a zero lower bound without compatibility. These are *state spreads*, not instrumental error or a disturbance formula.

2.11.2 Limiting cases. A momentum eigenstate is a plane wave $e^{ipx/\hbar}/\sqrt{2\pi\hbar}$, fully delocalized; a position eigenstate has momentum amplitude $e^{-ipx_0/\hbar}/\sqrt{2\pi\hbar}$, constant in magnitude. Both are nonnormalizable idealizations: do not substitute “ $0 \times \infty$ ” into the variance inequality. Interpret them as limits of normalized packets. For the source’s real centered Gaussian ($x_0 > 0$ is a *width*, not its center),

$$\psi(x) = \frac{e^{-x^2/(2x_0^2)}}{\pi^{1/4}\sqrt{x_0}}, \quad \phi(p) = \frac{\sqrt{x_0}}{\pi^{1/4}\sqrt{\hbar}} e^{-p^2 x_0^2/(2\hbar^2)}, \quad \Delta x = \frac{x_0}{\sqrt{2}}, \quad \Delta p = \frac{\hbar}{\sqrt{2}x_0}.$$

Square the amplitude *before* identifying a normal-distribution variance. This unchirped Gaussian saturates $\hbar/2$; a Gaussian with a quadratic phase need not saturate this product.

2.12 Worked examples

Source PDF pp. 38–39.

2.12.1 General Hermitian 2×2 matrix. For real $a_0, \mathbf{a}$,

$$X = a_0 I + \mathbf{a} \cdot \boldsymbol{\sigma} = \begin{pmatrix} a_0 + a_z & a_x - ia_y \\ a_x + ia_y & a_0 - a_z \end{pmatrix}, \quad \det(X - \lambda I) = (a_0 - \lambda)^2 - |\mathbf{a}|^2.$$

Thus $\lambda_{\pm} = a_0 \pm |\mathbf{a}|$. Read $a_y = -\text{Im } X_{12}$, not the opposite sign. Alternatively $(\mathbf{a} \cdot \boldsymbol{\sigma})^2 = |\mathbf{a}|^2 I$. When $|\mathbf{a}| > 0$, projectors $P_{\pm} = (I \pm \hat{\mathbf{a}} \cdot \boldsymbol{\sigma})/2$ give $X = \lambda_+ P_+ + \lambda_- P_-$. When $\mathbf{a} = 0$, $X = a_0 I$ is fully degenerate and $\hat{\mathbf{a}}$ is undefined; any orthonormal basis works.

2.12.2 Distributional matrix elements: source sign correction. With the stated Fourier convention the correct kernels are

$$\boxed{\langle p|x|p'\rangle = i\hbar\partial_p\delta(p-p') = -i\hbar\partial_{p'}\delta(p-p'), \quad \langle x|p|x'\rangle = -i\hbar\partial_x\delta(x-x')}.$$

The original PDF’s Eqs. (2.172)–(2.173) have both signs reversed. Indeed differentiate $e^{i(p'-p)x/\hbar}$ with respect to p under $\int dx/(2\pi\hbar)$, or integrate the kernel against $\phi(p')$ to obtain $i\hbar\phi'(p)$. Then

$$\langle p|[x, p]|p'\rangle = (p' - p)i\hbar\partial_p\delta(p-p') = i\hbar\delta(p-p').$$

Always specify which variable is differentiated; integration against a test function is the safest sign check.

2.13 Problems: solution roadmaps and final checks

Source PDF pp. 39–40.

1. **Spin spectral decomposition and phases.** Insert $I = |+\rangle\langle+| + |-\rangle\langle-|$ into $S_z I$ and apply each eigenvalue equation to obtain $S_z = (\hbar/2)(|+\rangle\langle+| - |-\rangle\langle-|)$. Equal probabilities and orthogonality allow

$$|x+\rangle = \frac{|+\rangle + e^{i\chi}|-\rangle}{\sqrt{2}}, \quad |x-\rangle = \frac{|+\rangle - e^{i\chi}|-\rangle}{\sqrt{2}},$$

up to independent overall phases. Setting $\chi = 0$ chooses the usual x axis/relative phase of the z basis. *The probabilities and orthogonality alone do not determine χ in a fixed z -basis convention*; the problem’s “up to an overall phase” needs this extra convention. Check zero overlap and unit norm explicitly.

2. **Cauchy–Schwarz and minimum uncertainty.** For $\beta \neq 0$, minimize $\|\alpha + \lambda\beta\|^2$ with

$$\lambda = -\frac{\langle\beta|\alpha\rangle}{\langle\beta|\beta\rangle} \Rightarrow |\langle\beta|\alpha\rangle|^2 \leq \langle\alpha|\alpha\rangle\langle\beta|\beta\rangle.$$

Equality iff the vectors are linearly dependent (including the trivial zero-vector case). Apply to fluctuation vectors and follow 2.11’s commutator split. For $d > 0$, write $\bar{x} = \langle x \rangle$, $\bar{p} = \langle p \rangle$ in

$$\psi(x) = (2\pi d^2)^{-1/4} \exp\left[\frac{i\bar{p}x}{\hbar} - \frac{(x - \bar{x})^2}{4d^2}\right].$$

$|\psi|^2$ is a normal density with mean $\bar{x}$ and variance d^2 . Differentiate once:

$$(p - \bar{p})\psi = \frac{i\hbar(x - \bar{x})}{2d^2}\psi, \quad \|(p - \bar{p})\psi\|^2 = \frac{\hbar^2}{4d^2}.$$

Hence $\Delta x = d$, $\Delta p = \hbar/(2d)$, product $\hbar/2$. The proportionality is imaginary, so covariance vanishes as required.

3. **Simultaneous diagonalization with degeneracy.** For the given $A = \text{diag}(a, -a, -a)$ and $B = b \oplus b\sigma_y$ in $(|1\rangle, |2\rangle, |3\rangle)$, the common orthonormal basis and eigenvalue pairs are

ket	A	B
$ 1\rangle$	a	b
$(2\rangle + i 3\rangle)/\sqrt{2}$	$-a$	b
$(2\rangle - i 3\rangle)/\sqrt{2}$	$-a$	$-b$

For $b \neq 0$, B has eigenvalue b twice and $-b$ once; for $b = 0$ all three are zero. $[A, B] = 0$ because the only off-diagonal B entries lie within the equal- $-a$ block. For $a \neq 0$, A has a twofold $-a$ eigenspace; if $a = 0$, it is fully degenerate. The displayed basis works even when a or b vanishes. Degeneracy gives freedom to diagonalize the other operator, not an obstruction.

4. **Optional: rotation of a Pauli vector.** Require real unit $\mathbf{n}$. Since $(\mathbf{n} \cdot \boldsymbol{\sigma})^2 = I$,

$$U(\varphi) = \cos(\varphi/2)I - i\sin(\varphi/2)\mathbf{n} \cdot \boldsymbol{\sigma}.$$

Expand $U(\mathbf{a} \cdot \boldsymbol{\sigma})U^\dagger$ using the Pauli product identity; collect coefficients to obtain the active right-handed Rodrigues rotation

$$\mathbf{a}' = \mathbf{a} \cos \varphi + (\mathbf{n} \times \mathbf{a}) \sin \varphi + \mathbf{n}(\mathbf{n} \cdot \mathbf{a})(1 - \cos \varphi).$$

For $\mathbf{n} = \hat{\mathbf{z}}$, $U = \text{diag}(e^{-i\varphi/2}, e^{i\varphi/2})$ and $(a'_x, a'_y, a'_z) = (a_x \cos \varphi - a_y \sin \varphi, a_x \sin \varphi + a_y \cos \varphi, a_z)$. Check $U\sigma_x U^\dagger = \cos \varphi \sigma_x + \sin \varphi \sigma_y$. Reversing the conjugation reverses the angle. $U(2\pi) = -I$ leaves the physical ray and the rotated Pauli vector unchanged.

Self-checks (answers below). (i) Can increasing intensity overcome the single-photon threshold? (ii) What fraction of a prepared $z+$ beam survives $x+$ then $z-$? (iii) Is $(1, i)^T/\sqrt{2}$ an $x+$ state? (iv) Does $[A, B] \neq 0$ forbid every common eigenket? (v) What happens to $\psi(x)$ under $T(a)$? (vi) Is the source’s Gaussian width x_0 its standard deviation? (vii) Which kernel sign produces $x = i\hbar\partial_p$?

Self-check answers. (i) No; $\hbar\nu \geq \Phi$ is required in that regime. (ii) 1/4 jointly, 1/2 conditional on passing $x+$. (iii) No: it is $y+$, with $P(x\pm) = 1/2$. (iv) No; it forbids a complete common eigenbasis in the stated setting. (v) $\psi(x - a)$, a rightward shift by a . (vi) No: $\Delta x = x_0/\sqrt{2}$. (vii) $+i\hbar\partial_p\delta(p - p')$, equivalently $-i\hbar\partial_{p'}\delta(p - p')$.

Chapter 3: Quantum Dynamics

Scope. Original extract pp. 41–57 covers Sections 3.1–3.4; the supplied continuation pp. 58–74 completes Sections 3.4.1–3.9. The continuation is kept in `chapter3_continuation.tex` so its source boundary remains explicit.

3.1 Time evolution and the Schrödinger equation

Source PDF pp. 41–46.

3.1.1 The time-evolution operator

Setting: closed-system, continuous unitary dynamics; time is a parameter, not a position-like observable. With a common Hamiltonian governing every initial state,

$$\begin{aligned} |\psi(t)\rangle &= U(t, t_0)|\psi(t_0)\rangle, & U^\dagger U &= U U^\dagger = 1, \\ U(t_0, t_0) &= 1, & U(t_2, t_0) &= U(t_2, t_1)U(t_1, t_0), & U^{-1}(t, t_0) &= U(t_0, t) = U^\dagger(t, t_0). \end{aligned}$$

Derivation backbone: continuity gives $U(t + \delta t, t) = 1 - i\Omega(t)\delta t + O(\delta t^2)$. Then $U^\dagger U = 1 + i(\Omega^\dagger - \Omega)\delta t + O(\delta t^2)$ implies $\Omega^\dagger = \Omega$. Identifying the energy generator as $H = \hbar\Omega$ gives

$$U(t + \delta t, t) = 1 - \frac{i}{\hbar}H(t)\delta t + O(\delta t^2).$$

Energy generates time translations, just as momentum generates spatial translations: $T(\delta \mathbf{r}) = 1 - i\delta \mathbf{r} \cdot \mathbf{p}/\hbar + O(\delta r^2)$. Unitarity establishes a Hermitian generator; its identification with the physical energy is additional physical input.

3.1.2 Derivation of the Schrödinger equation

Insert the infinitesimal operator into the composition law, subtract $U(t, t_0)$ and divide by δt :

$$i\hbar\partial_t U(t, t_0) = H(t)U(t, t_0), \quad i\hbar\partial_t |\psi(t)\rangle = H(t)|\psi(t)\rangle.$$

For a local scalar potential and no vector potential, $H = \mathbf{p}^2/(2m) + V(\mathbf{r}, t)$, $\mathbf{p} = -i\hbar\nabla$, so

$$i\hbar\partial_t \psi(\mathbf{r}, t) = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r}, t) \right] \psi(\mathbf{r}, t).$$

First order in time: one initial state, not an independent initial time derivative, determines the evolution (with the Hamiltonian and spatial domain/boundary conditions specified).

3.1.3 Solutions for the time-evolution operator

Time-independent H : $U = e^{-iH(t-t_0)/\hbar}$; the exponential is an operator power series, not division by H .

Commuting time-dependent H : if $[H(t_1), H(t_2)] = 0$ for every pair of times,

$$U(t, t_0) = \exp \left[-\frac{i}{\hbar} \int_{t_0}^t H(s) ds \right].$$

General noncommuting case (write the following for $t \geq t_0$): integrate the evolution equation and iterate,

$$\begin{aligned} U(t, t_0) &= 1 - \frac{i}{\hbar} \int_{t_0}^t dt_1 H(t_1)U(t_1, t_0) \\ &= 1 - \frac{i}{\hbar} \int_{t_0}^t dt_1 H(t_1) + \left(-\frac{i}{\hbar} \right)^2 \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 H(t_1)H(t_2) + \cdots \\ &= \mathcal{T} \exp \left[-\frac{i}{\hbar} \int_{t_0}^t H(s) ds \right]. \end{aligned}$$

$\mathcal{T}$ puts later-time factors on the left. The ordered triangular integral has no extra $1/2!$. Equivalently, integrate the time-ordered product over the full square and include a factor $1/2!$. A magnetic field whose direction changes can give noncommuting spin Hamiltonians. **Pitfall:** a Hermitian $H(t)$ does not imply that Hamiltonians at different times commute; dropping $\mathcal{T}$ is generally wrong.

3.1.4 Energy eigenstates and stationary states

For time-independent H , use a complete orthonormal energy basis, with extra degeneracy labels when needed (and integrals for continuous spectra):

$$H|n\rangle = E_n|n\rangle, \quad \sum_n |n\rangle\langle n| = 1,$$

$$U(t, t_0) = \sum_n e^{-iE_n(t-t_0)/\hbar} |n\rangle\langle n|, \quad |\psi(t)\rangle = \sum_n c_n e^{-iE_n(t-t_0)/\hbar} |n\rangle.$$

The populations $|c_n|^2$ stay fixed; relative phases evolve. To locate measurable oscillations, expand a time-independent observable:

$$\langle A \rangle_t = \sum_{m,n} c_m^* c_n A_{mn} e^{i(E_m - E_n)(t-t_0)/\hbar}, \quad A_{mn} = \langle m|A|n\rangle.$$

Thus energy differences determine frequencies, while coherences and matrix elements determine which frequencies appear. A superposition need not make every observable oscillate.

Stationarity: an energy eigenstate acquires only a global phase, so all time-independent observables have time-independent probability distributions. This is not the assertion that the particle is at rest. Any superposition wholly within one degenerate energy eigenspace is also stationary. Explicitly time-dependent observables are excluded from this statement.

Degeneracy and quantum numbers: a complete set of commuting observables satisfies

$$[A_i, A_j] = 0, \quad [A_i, H] = 0, \quad H|E, a, b, \dots\rangle = E|E, a, b, \dots\rangle.$$

Their simultaneous eigenvalues distinguish states with the same energy; enough independent commuting labels are needed for a complete labeling. Conservation also requires no explicit time dependence of these observables.

3.1.5 Example: spin precession

Keep the source's signed convention, rather than inserting an electron-charge sign by memory:

$$H = -\boldsymbol{\mu} \cdot \mathbf{B} = -\gamma \mathbf{S} \cdot \mathbf{B}, \quad \mathbf{B} = B\hat{\mathbf{z}}, \quad \omega = -\gamma B, \quad H = \omega S_z.$$

For spin 1/2, $S_z = \frac{\hbar}{2} \text{diag}(1, -1)$ and $E_{\pm} = \pm \hbar\omega/2$. Therefore

$$c_+|+z\rangle + c_-|-z\rangle \mapsto c_+e^{-i\omega t/2}|+z\rangle + c_-e^{i\omega t/2}|-z\rangle.$$

Starting from $|+x\rangle = (|+z\rangle + |-z\rangle)/\sqrt{2}$, project onto $|\pm x\rangle = (|+z\rangle \pm |-z\rangle)/\sqrt{2}$:

$$\langle +x|\psi(t)\rangle = \cos(\omega t/2), \quad \langle -x|\psi(t)\rangle = -i \sin(\omega t/2),$$

$$P(S_x = +\hbar/2) = \cos^2(\omega t/2), \quad P(S_x = -\hbar/2) = \sin^2(\omega t/2),$$

$$\langle S_x \rangle_t = \frac{\hbar}{2} \cos \omega t, \quad \langle S_y \rangle_t = \frac{\hbar}{2} \sin \omega t, \quad \langle S_z \rangle_t = 0.$$

Mechanism: the two S_z components retain their weights and develop a relative phase. If the initial state is $|+z\rangle$, there is instead only a global phase and no precessing transverse mean. **Sign check:** with $H = \omega S_z$, $\dot{S}_x = (i/\hbar)[\omega S_z, S_x] = -\omega S_y$ and $\dot{S}_y = \omega S_x$, consistent with the displayed trajectory. The sign of γ determines its sense.

3.2 The Schrödinger and Heisenberg pictures

Source PDF pp. 47–50.

3.2.1 Unitary changes of representation

$U^\dagger U = 1$ preserves inner products. The identity $(\langle\alpha|U^\dagger)X(U|\beta\rangle) = \langle\alpha|(U^\dagger XU)|\beta\rangle$ lets us move evolution from states to operators without changing matrix elements. A change of picture changes representation, not experimental predictions.

3.2.2 Two equivalent pictures

Using the same reference time t_0 and propagator $U(t, t_0)$,

$$\begin{aligned} |\psi_S(t)\rangle &= U|\psi_S(t_0)\rangle, & |\psi_H\rangle &= U^\dagger|\psi_S(t)\rangle = |\psi_S(t_0)\rangle, \\ X_H(t) &= U^\dagger X_S(t) U, & \langle\psi_S(t)|X_S(t)|\psi_S(t)\rangle &= \langle\psi_H|X_H(t)|\psi_H\rangle. \end{aligned}$$

Schrödinger operators may have *explicit* time dependence; they are fixed only when that dependence is absent. Transform all operators consistently: $[A_H, B_H] = U^\dagger[A_S, B_S]U$.

3.2.3 The equation of motion in the Heisenberg picture

Differentiate all three factors of $U^\dagger X_S U$, using $\dot{U} = -iH_S U/\hbar$ and $\dot{U}^\dagger = iU^\dagger H_S/\hbar$:

$$\boxed{\frac{dX_H}{dt} = \frac{i}{\hbar}[H_H, X_H] + U^\dagger \frac{\partial X_S}{\partial t} U}, \quad H_H = U^\dagger H_S U.$$

Equivalently, $i\hbar\dot{X}_H = [X_H, H_H] + i\hbar(\partial_t X_S)_H$. For time-independent H , $H_H = H_S$; do not assume this for arbitrary time-dependent noncommuting $H_S(t)$. **Conservation:** $[X_S, H_S] = 0$ together with $\partial_t X_S = 0$ implies $\dot{X}_H = 0$. Commutation alone is insufficient if X_S explicitly changes with time. In particular $d\langle H\rangle/dt = \langle\partial_t H\rangle$.

3.2.4 Constructing the Hamiltonian and the classical correspondence

Model assumptions: for a nonrelativistic particle in a scalar potential, $H = \mathbf{p}^2/(2m) + V(\mathbf{x})$ and $[x_i, p_j] = i\hbar\delta_{ij}$. Promoting classical variables to operators is not a universal quantization algorithm: operator ordering can be ambiguous; spin is an internal degree of freedom; symmetry and gauge invariance constrain interactions; closed-system evolution requires a Hermitian Hamiltonian. For suitably expandable functions,

$$[x_i, F(\mathbf{p})] = i\hbar \frac{\partial F}{\partial p_i}, \quad [p_i, G(\mathbf{x})] = -i\hbar \frac{\partial G}{\partial x_i}.$$

Free particle: $\dot{\mathbf{p}} = 0$, $\dot{\mathbf{x}} = \mathbf{p}/m$ imply

$$\mathbf{x}(t) = \mathbf{x}(0) + \frac{t}{m}\mathbf{p}(0), \quad [x_i(t), x_j(0)] = -\frac{i\hbar t}{m}\delta_{ij}.$$

Equal-time position components commute; different-time positions need not. **Scalar potential and Ehrenfest:**

$$\begin{aligned} \dot{x}_i &= p_i/m, & \dot{p}_i &= -\partial_i V(\mathbf{x}), & m\ddot{x}_i &= -\partial_i V(\mathbf{x}), \\ \boxed{\frac{d\langle x_i \rangle}{dt} &= \frac{\langle p_i \rangle}{m}, & \frac{d\langle p_i \rangle}{dt} &= -\langle \partial_i V(\mathbf{x}) \rangle.} \end{aligned}$$

Classical limit pitfall: generally $\langle\partial_i V(\mathbf{x})\rangle \neq \partial_i V(\langle\mathbf{x}\rangle)$. Equality is exact for potentials at most quadratic, whose force is linear/constant; it is approximate for a narrow packet in a slowly varying force. Exact classical motion of the mean does not imply a classical probability distribution or zero fluctuations.

3.3 The simple harmonic oscillator

Source PDF pp. 50–56.

Why this model: near a smooth stable minimum x_{eq} with $V''(x_{\text{eq}}) > 0$, $V(x) \simeq V(x_{\text{eq}}) + \frac{1}{2}V''(x_{\text{eq}})(x - x_{\text{eq}})^2$. Shift the origin and discard the constant energy offset; $m\omega^2 = V''(x_{\text{eq}})$, $\omega > 0$. The approximation needs sufficiently small excursions; a flat minimum need not be harmonic. Classically $x = A \cos(\omega t + \delta)$, $p = -m\omega A \sin(\omega t + \delta)$ and $E = m\omega^2 A^2/2$.

3.3.1 Ladder operators and the energy spectrum

Start from $H = p^2/(2m) + m\omega^2 x^2/2$, $[x, p] = i\hbar$. Then $[x, H] = i\hbar p/m$, $[p, H] = -i\hbar m\omega^2 x$, so $[x \pm ip/(m\omega), H] = \pm\hbar\omega[x \pm ip/(m\omega)]$. Define

$$a = \sqrt{\frac{m\omega}{2\hbar}} \left(x + \frac{ip}{m\omega} \right), \quad a^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(x - \frac{ip}{m\omega} \right), \quad [a, a^\dagger] = 1,$$

$$x = \sqrt{\frac{\hbar}{2m\omega}}(a + a^\dagger), \quad p = -i\sqrt{\frac{m\hbar\omega}{2}}(a - a^\dagger), \quad \boxed{H = \hbar\omega(N + \tfrac{1}{2}), \quad N = a^\dagger a.}$$

Spectrum proof: $\langle N \rangle = \|a|\psi\rangle\|^2 \geq 0$ and $[N, a] = -a$, $[N, a^\dagger] = a^\dagger$. Repeated lowering of an eigenstate must terminate before a negative eigenvalue; the terminating state has norm $\|a|0\rangle\|^2 = 0$, hence eigenvalue zero. The original eigenvalue is therefore an integer. Norms fix the ladder coefficients:

$$a|n\rangle = \sqrt{n}|n-1\rangle, \quad a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle, \quad |n\rangle = \frac{(a^\dagger)^n}{\sqrt{n!}}|0\rangle,$$

$$\boxed{E_n = \hbar\omega(n + \tfrac{1}{2}), \quad n = 0, 1, \dots, \quad a|0\rangle = 0.}$$

The vacuum is not the zero vector. Zero-point energy $E_0 = \hbar\omega/2$ reflects the impossibility of simultaneously sharp $x = p = 0$.

3.3.2 Matrix elements of position and momentum

$$\langle n'|x|n\rangle = \sqrt{\frac{\hbar}{2m\omega}}(\sqrt{n}\delta_{n',n-1} + \sqrt{n+1}\delta_{n',n+1}),$$

$$\langle n'|p|n\rangle = i\sqrt{\frac{m\hbar\omega}{2}}(-\sqrt{n}\delta_{n',n-1} + \sqrt{n+1}\delta_{n',n+1}).$$

In the ordered basis $|0\rangle, |1\rangle, \dots$, $a_{n',n} = \sqrt{n}\delta_{n',n-1}$: its first upper off-diagonal is 1, $\sqrt{2}, \sqrt{3}, \dots$; $a^\dagger$ is its conjugate transpose. Linear x or p couplings connect only $\Delta n = \pm 1$; this is not a selection rule for arbitrary nonlinear couplings.

3.3.3 Wave functions and Hermite polynomials

Set $x_0 = \sqrt{\hbar/(m\omega)}$, $\xi = x/x_0$ (here x_0 is a length, not the equilibrium location). Then $a = (\xi + \partial_\xi)/\sqrt{2}$ and $a\psi_0 = 0$ gives $\psi_0 \propto e^{-\xi^2/2}$; normalize in dx , then apply $(a^\dagger)^n/\sqrt{n!}$:

$$\boxed{\psi_n(x) = \frac{H_n(x/x_0)e^{-x^2/(2x_0^2)}}{\pi^{1/4}\sqrt{2^n n!}x_0}.$$

H_n are the physicists' Hermite polynomials. There are n nodes and parity $(-1)^n$; the Gaussian factor uses the same x_0 for all n . Using ladder algebra, $\langle x \rangle_n = \langle p \rangle_n = 0$ and

$$\langle x^2 \rangle_n = \frac{\hbar}{m\omega}(n + \tfrac{1}{2}), \quad \langle p^2 \rangle_n = m\hbar\omega(n + \tfrac{1}{2}), \quad \Delta x \Delta p = \hbar(n + \tfrac{1}{2}).$$

Only the ground number state saturates $\Delta x \Delta p = \hbar/2$; its rms width is $x_0/\sqrt{2}$, not x_0 .

3.3.4 Time evolution of oscillator observables

Heisenberg operators (reference time zero) satisfy

$$\dot{x} = p/m, \quad \dot{p} = -m\omega^2 x, \quad \dot{a} = -i\omega a, \quad \dot{a}^\dagger = i\omega a^\dagger.$$

Solve either the coupled equations or the ladder equations:

$$\begin{aligned} a(t) &= a(0)e^{-i\omega t}, & a^\dagger(t) &= a^\dagger(0)e^{i\omega t}, \\ \boxed{x(t) &= x(0) \cos \omega t + \frac{p(0)}{m\omega} \sin \omega t}, \\ \boxed{p(t) &= -m\omega x(0) \sin \omega t + p(0) \cos \omega t}. \end{aligned}$$

These are operator identities, valid in every state. They do not say every state has an oscillating mean: $\langle n|x(t)|n \rangle = \langle n|p(t)|n \rangle = 0$.

For a normalized $|\psi\rangle = c_0|0\rangle + c_1|1\rangle$ at $t = 0$, the off-diagonal matrix elements alone survive:

$$\begin{aligned} \langle x(t) \rangle &= \sqrt{\frac{\hbar}{2m\omega}} (c_0^* c_1 e^{-i\omega t} + c_1^* c_0 e^{i\omega t}) \\ &= 2\sqrt{\frac{\hbar}{2m\omega}} |c_0 c_1| \cos(\omega t - \delta), \quad \delta = \arg(c_0^* c_1). \end{aligned}$$

The mean oscillation is a coherence effect between neighboring levels. If one coefficient vanishes, the amplitude vanishes and δ is irrelevant. A statistical mixture with the same populations lacks this cross term.

Figure/correspondence lesson: a highly excited number state has a rapidly oscillating spatial probability density, whose coarse-grained envelope approaches the classical distribution at the same energy. It remains stationary: large n alone does not produce a localized packet traveling around an orbit.

3.3.5 Coherent states

Define $a|\alpha\rangle = \alpha|\alpha\rangle$, $\alpha \in \mathbb{C}$; a is not Hermitian, so its eigenvalue need not be real. Write $|\alpha\rangle = \sum_n c_n |n\rangle$. Equating coefficients gives $c_{n+1}\sqrt{n+1} = \alpha c_n$, hence $c_n = c_0 \alpha^n / \sqrt{n!}$; normalization gives $|c_0| = e^{-|\alpha|^2/2}$.

$$\begin{aligned} \boxed{|\alpha\rangle &= e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle}, \\ P_n &= e^{-|\alpha|^2} \frac{|\alpha|^{2n}}{n!}, \quad \bar{n} = \langle N \rangle = |\alpha|^2, \quad \langle H \rangle = \hbar\omega(|\alpha|^2 + \tfrac{1}{2}). \end{aligned}$$

The occupation distribution is Poisson. **Phase matters:** $|\alpha| = \sqrt{\bar{n}}$ always, whereas $\alpha = \sqrt{\bar{n}}$ requires *nonnegative* real α . Merely saying “real” (as in the source) misses the negative-real possibility.

Displacement construction:

$$|\alpha\rangle = D(\alpha)|0\rangle, \quad D(\alpha) = \exp(\alpha a^\dagger - \alpha^* a).$$

The exponent is anti-Hermitian, so D is unitary. Real and imaginary parts set the position and momentum displacements, respectively. Multiplying each number-state coefficient by its energy phase gives

$$\boxed{|\alpha, t\rangle = e^{-i\omega t/2} |\alpha e^{-i\omega t}\rangle}.$$

For $\alpha_t = \alpha e^{-i\omega t}$, ladder algebra yields

$$\begin{aligned} \langle x \rangle_t &= \sqrt{\frac{2\hbar}{m\omega}} \operatorname{Re} \alpha_t, & \langle p \rangle_t &= \sqrt{2m\hbar\omega} \operatorname{Im} \alpha_t, \\ (\Delta x)^2 &= \frac{\hbar}{2m\omega}, & (\Delta p)^2 &= \frac{m\hbar\omega}{2}, \quad \Delta x \Delta p = \frac{\hbar}{2}. \end{aligned}$$

The packet keeps its shape and its center follows a classical orbit, while nonzero vacuum fluctuations remain. The global factor $e^{-i\omega t/2}$ changes no probabilities. Coherent states are special superpositions, not energy eigenstates except for $\alpha = 0$.

3.4 Coordinate-space solution of the harmonic oscillator

Source PDF pp. 56–57.

Boundary-value problem: on the whole real line, require a square-integrable wave function solving

$$\left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2} m \omega^2 x^2 \right] \psi = E \psi.$$

Nondimensionalization: with $x_0 = \sqrt{\hbar/(m\omega)}$, $\xi = x/x_0$ and $\lambda = 2E/(\hbar\omega)$,

$$(-\partial_\xi^2 + \xi^2)\psi = \lambda\psi.$$

Asymptotics: for $|\xi| \gg 1$, the leading exponential behaviors are $e^{\pm\xi^2/2}$; normalizability excludes the growing branch. The allowed solution can have an additional polynomial factor, so do not impose a pure Gaussian for every level. Put $\psi = H(\xi)e^{-\xi^2/2}$; then

$$\begin{aligned} \psi'' &= e^{-\xi^2/2} [H'' - 2\xi H' + (\xi^2 - 1)H], \\ H'' - 2\xi H' + (\lambda - 1)H &= 0. \end{aligned}$$

Power series and termination: insert $H = \sum_{r=0}^{\infty} a_r \xi^r$ and align powers:

$$(r+2)(r+1)a_{r+2} + (\lambda - 2r - 1)a_r = 0, \quad \boxed{a_{r+2} = \frac{2r+1-\lambda}{(r+2)(r+1)} a_r.}$$

Even and odd coefficients are separate chains, seeded by a_0 and a_1 . A nonterminating chain generically supplies an e^{ξ^2} factor in H , hence a nonnormalizable $e^{+\xi^2/2}$ branch in ψ ; decay at only one end is insufficient. The acceptable full-line solution is a terminating parity chain; the other chain is set to zero. Termination at degree n requires

$$\boxed{\lambda = 2n + 1 \implies E_n = \hbar\omega(n + \tfrac{1}{2}), \quad n = 0, 1, \dots}$$

The surviving polynomial solves the Hermite equation,

$$H_n'' - 2\xi H_n' + 2nH_n = 0, \quad H_0 = 1, \quad H_1 = 2\xi, \quad H_2 = 4\xi^2 - 2.$$

With the standard Hermite normalization one recovers the normalized $\psi_n(x)$ in 3.3.3. **Exam logic:** the differential equation admits formal solutions at other energies; acceptable boundary conditions select the discrete spectrum. Nondimensionalization alone does not quantize energy.

Self-test for 3.1–3.4: reconstruct before looking at the answers

1. Why is the infinitesimal generator Hermitian? When may time ordering be removed?
2. Must a superposition evolve observably? What happens within one degenerate eigenspace?
3. For $H = \omega S_z$ and initial $|+x\rangle$, find $P(-x)$ and the sign of $\langle S_y \rangle$.
4. State the full Heisenberg equation and the two sufficient conditions for conservation.
5. Derive $[x(t), x(0)]$ for a free particle. Why does Ehrenfest not generally give one classical trajectory?
6. What forces integer oscillator occupation, and what are $\langle 2|x|0\rangle$ and $\Delta x \Delta p$ in $|2\rangle$?
7. Does a highly excited number state oscillate? What ingredient makes $c_0|0\rangle + c_1|1\rangle$ oscillate?
8. If $\alpha = i\beta$ with real $\beta > 0$, give $\bar{n}$, $\langle x(0) \rangle$, $\langle p(0) \rangle$, and the evolved coherent label.
9. At $\lambda = 5$, which series seed must vanish, what is a_2/a_0 , and what energy follows?

Compact answers. **1.** The $O(\delta t)$ term of $U^\dagger U - 1$ is $i(\Omega^\dagger - \Omega)\delta t$; pairwise commuting $H(t)$ permits dropping $\mathcal{T}$. **2.** No: observable oscillations require nonzero cross terms with unequal energies; one degenerate eigenspace acquires one global phase. **3.** $\sin^2(\omega t/2)$; $\langle S_y \rangle = (\hbar/2) \sin \omega t$ with the signed ω . **4.** $\dot{X}_H = (i/\hbar)[H_H, X_H] + (\partial_t X_S)_H$; zero commutator and zero explicit derivative. **5.** $[x(0) + tp(0)/m, x(0)] = -i\hbar t/m$; generally $\langle V'(x) \rangle \neq V'(\langle x \rangle)$. **6.** Positivity of N forces lowering to terminate at zero; 0 and $5\hbar/2$. **7.** Its mean does not; neighboring-level coherence $c_0^*c_1 \neq 0$ produces the oscillation. **8.** β^2 , 0, $\sqrt{2m\hbar\omega}\beta$, and $i\beta e^{-i\omega t}$ (plus the state's global phase). **9.** $a_1 = 0$, $a_2/a_0 = -2$, $E = 5\hbar\omega/2$; $a_4 = 0$ and $H \propto H_2$.

3.4.1 A related example: the linear potential $V(x) = k|x|$

Source continuation PDF p. 58.

For $k > 0$ on the full line,

$$-\frac{\hbar^2}{2m}\psi'' + k|x|\psi = E\psi, \quad x_0 = \left(\frac{\hbar^2}{2mk}\right)^{1/3}, \quad y = \frac{x}{x_0}, \quad \epsilon = \frac{E}{kx_0}.$$

The scale follows by balancing $\hbar^2/(2mx_0^2) = kx_0$, so $E \sim kx_0 \propto \hbar^{2/3}k^{2/3}m^{-1/3}$. The consistent dimensionless equation is

$$-\frac{d^2\psi}{dy^2} + |y|\psi = \epsilon\psi.$$

Source correction: the continuation PDF's Eq. (3.153) instead prints $d^2\psi/dy^2 - 2|y|\psi = -2\epsilon\psi$. Its factors of two conflict with the stated x_0 , ϵ , and ground energy. For $y > 0$, normalizability leaves $\psi \propto \text{Ai}(y - \epsilon)$ and excludes Bi. Parity imposes

$$\text{Ai}'(-\epsilon) = 0 \quad (\text{even}), \quad \text{Ai}(-\epsilon) = 0 \quad (\text{odd}).$$

The first zero of Ai' gives $E_0 = 1.018792 \dots kx_0$. Both ψ and ψ' are continuous at $x = 0$: V has a cusp but no delta function. Dimensional analysis fixes the scale, not the Airy-zero coefficient.

3.5 The wave function, probability current, and classical limit

Source continuation PDF pp. 58–60.

3.5.1 Probability density and continuity equation

For $H = -\hbar^2\nabla^2/(2m) + V(\mathbf{r}, t)$ with real V , set $\rho = |\psi|^2$. Multiply the Schrödinger equation by ψ^* , its conjugate by ψ , and subtract:

$$\begin{aligned} \partial_t \rho &= \frac{i\hbar}{2m}(\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*) = -\nabla \cdot \mathbf{j}, \\ \mathbf{j} &= \left[\frac{\hbar}{m} \text{Im}(\psi^* \nabla \psi) = \frac{1}{m} \text{Re}[\psi^* (-i\hbar \nabla) \psi] \right]. \end{aligned}$$

Thus $\partial_t \rho + \nabla \cdot \mathbf{j} = 0$ and $d \int_{\Omega} \rho d^3r / dt = -\oint_{\partial\Omega} \mathbf{j} \cdot d\mathbf{S}$. Total probability is conserved when the boundary flux vanishes. For an effective complex potential $V = V_R + iV_I$,

$$\partial_t \rho + \nabla \cdot \mathbf{j} = \frac{2V_I}{\hbar} \rho;$$

$V_I < 0$ is a sink. This phenomenologically removes unresolved channels and is not unitary closed-system dynamics. The displayed $\mathbf{j}$ is probability current; electric current is $q\mathbf{j}$.

3.5.2 Phase and probability flow

Where $\rho > 0$, write

$$\psi = \sqrt{\rho} e^{iS/\hbar}, \quad \mathbf{j} = \rho \frac{\nabla S}{m} \equiv \rho \mathbf{v}.$$

Phase gradients drive flow. The field $\mathbf{v}$ is not automatically the trajectory velocity of an individual particle. At nodes this polar form is singular and S may be multivalued.

3.5.3 Quantum Hamilton–Jacobi equation and classical limit

Substitution of $\sqrt{\rho} e^{iS/\hbar}$ and separation into imaginary and real parts gives the exact pair

$$\begin{aligned} \partial_t \rho + \nabla \cdot \left(\rho \frac{\nabla S}{m} \right) &= 0, \\ \partial_t S + \frac{|\nabla S|^2}{2m} + V + Q &= 0, \quad Q = -\frac{\hbar^2}{2m} \frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}}. \end{aligned}$$

The first transports amplitude; the second transports phase. Classical Hamilton–Jacobi theory follows where Q (and its relevant gradients) is negligible relative to the other scales. This is a slow-amplitude/WKB condition, not merely the slogan $\hbar \rightarrow 0$; it can fail near nodes, turning points, sharp boundaries, and interference fringes. For a stationary state, $S(\mathbf{r}, t) = W(\mathbf{r}) - Et$.

3.6 Propagators and path integrals

Source continuation PDF pp. 60–65.

Unless stated otherwise, take one spatial dimension, $t > t_0$, and $H = p^2/(2m) + V(x)$ with real, sufficiently regular V .

3.6.1 Propagator: one kernel evolves every state

$$K(x, t; x', t_0) = \langle x | U(t, t_0) | x' \rangle, \quad \boxed{\psi(x, t) = \int dx' K(x, t; x', t_0) \psi(x', t_0).}$$

For time-independent H ,

$$K = \sum_n \psi_n(x) \psi_n^*(x') e^{-iE_n(t-t_0)/\hbar},$$

with continuum integrals added when required. Three operator identities become three reconstruction checks:

$$\begin{aligned} \lim_{t \rightarrow t_0^+} K(x, t; x', t_0) &= \delta(x - x') \quad \text{distributionally,} \\ K(x_2, t_2; x_0, t_0) &= \int dx_1 K(x_2, t_2; x_1, t_1) K(x_1, t_1; x_0, t_0), \\ i\hbar \partial_t K(x, t; x', t_0) &= H_x K(x, t; x', t_0). \end{aligned}$$

For $K_R = \Theta(t - t_0)K$,

$$(i\hbar \partial_t - H_x)K_R = i\hbar \delta(t - t_0) \delta(x - x').$$

Some texts instead define $G_R = -(i/\hbar)K_R$, whose right-hand side is simply $\delta(t - t_0)\delta(x - x')$; do not mix the two normalizations.

3.6.2 Free and harmonic-oscillator propagators

Insert momentum completeness, with $\langle x | p \rangle = e^{ipx/\hbar}/\sqrt{2\pi\hbar}$ and $T = t - t_0 > 0$:

$$\begin{aligned} K_0(x, t; x', t_0) &= \int \frac{dp}{2\pi\hbar} e^{\frac{i}{\hbar}[p(x-x') - p^2 T/(2m)]} \\ &= \boxed{\left(\frac{m}{2\pi i\hbar T}\right)^{1/2} e^{\frac{im(x-x')^2}{2\hbar T}}} = \left(\frac{m}{2\pi i\hbar T}\right)^{1/2} e^{iS_{\text{cl}}/\hbar}, \end{aligned}$$

where $S_{\text{cl}} = m(x - x')^2/(2T)$. The prefactor fixes dimensions, normalization, composition, and the delta-function limit; $e^{iS_{\text{cl}}/\hbar}$ alone is not the propagator. For comparison,

$$K_{\text{ho}}(x'', t; x', t_0) = \left(\frac{m\omega}{2\pi i\hbar \sin \omega T}\right)^{1/2} \exp\left\{\frac{im\omega}{2\hbar \sin \omega T}[(x''^2 + x'^2) \cos \omega T - 2x''x']\right\}.$$

Continue the square-root phase consistently through the caustics $\sin \omega T = 0$; the kernel there is distributional. The $\omega \rightarrow 0$ limit recovers K_0 .

3.6.3 Trace and spectral poles

For a discrete spectrum, with convergence/regulation understood,

$$G(t) = \int dx K(x, t; x, 0) = \text{Tr } e^{-iHt/\hbar} = \sum_n e^{-iE_n t/\hbar},$$

and

$$\boxed{\tilde{G}(E) = -\frac{i}{\hbar} \int_0^\infty dt e^{i(E+i0^+)t/\hbar} G(t) = \sum_n \frac{1}{E - E_n + i0^+}.}$$

Thus energy eigenvalues are poles. In infinite volume or with a continuum the trace may diverge and needs a box, density-of-states normalization, imaginary time, or another regulator.

3.6.4 Why K is an amplitude

In Heisenberg notation $K(x, t; x', t_0) = \langle x, t | x', t_0 \rangle$. It is an amplitude density, not a probability. Indistinguishable alternatives are added before taking $|\cdot|^2$; summing path probabilities would erase interference.

3.6.5 Time slicing: from composition to the path integral

Divide $T = t_f - t_i = N\varepsilon$, set $x_0 = x_i$, $x_N = x_f$, and insert $N - 1$ position identities:

$$K = \int \prod_{j=1}^{N-1} dx_j \prod_{j=0}^{N-1} \langle x_{j+1} | e^{-iH\varepsilon/\hbar} | x_j \rangle.$$

For $H = p^2/(2m) + V(x)$, a symmetric splitting (or equivalent consistent midpoint prescription) and a momentum Gaussian give

$$\begin{aligned} \langle x_{j+1} | e^{-iH\varepsilon/\hbar} | x_j \rangle &\simeq \left(\frac{m}{2\pi i \hbar \varepsilon} \right)^{1/2} \exp \left\{ \frac{i\varepsilon}{\hbar} \left[\frac{m}{2} \left(\frac{x_{j+1} - x_j}{\varepsilon} \right)^2 - V(\bar{x}_j) \right] \right\}, \\ K &= \lim_{N \rightarrow \infty} \left(\frac{m}{2\pi i \hbar \varepsilon} \right)^{N/2} \int \prod_{j=1}^{N-1} dx_j e^{iS_N/\hbar} \equiv \int_{x_i}^{x_f} \mathcal{D}x e^{iS[x]/\hbar}, \\ S[x] &= \int_{t_i}^{t_f} dt \left[\frac{1}{2} m \dot{x}^2 - V(x) \right]. \end{aligned}$$

This limit *defines* the formal $\mathcal{D}x$, including its normalization. The source first displays the endpoint-ordered first-order Trotter product and then invokes the usual symmetric midpoint prescription; these are equivalent in the continuum for a regular coordinate-only V , but midpoint discretization should be tied to symmetric splitting when ordering matters.

The integral sums amplitudes over all fixed-endpoint histories; it does not say the particle secretly takes one classical trajectory. When stationary phase is controlled, rapidly varying phases away from $\delta S = 0$ cancel while nearby paths add coherently. Include every stationary path and its fluctuation phase; for quadratic actions the Gaussian semiclassical evaluation is exact. **Pitfalls:** never omit the short-time normalization, promote a Trotter approximation to an exact finite- ε identity, or assume typical real-time paths are differentiable classical curves. Velocity-dependent/gauge couplings require a specified ordering and discretization.

3.6.6 Recovering the Schrödinger equation

For one short step put $\xi = x - x'$ and expand:

$$\begin{aligned} \psi(x, t + \varepsilon) &= \left(\frac{m}{2\pi i \hbar \varepsilon} \right)^{1/2} \int d\xi e^{im\xi^2/(2\hbar\varepsilon)} \left(1 - \frac{i\varepsilon}{\hbar} V(x) + \dots \right) \\ &\times \left(\psi - \xi \partial_x \psi + \frac{\xi^2}{2} \partial_x^2 \psi + \dots \right). \end{aligned}$$

The normalized oscillatory Gaussian has $\langle \xi \rangle = 0$ and $\langle \xi^2 \rangle = i\hbar\varepsilon/m$. Therefore $\xi \sim \sqrt{\varepsilon}$: the quadratic term is $O(\varepsilon)$ and must be retained. Comparing terms gives

$$\boxed{i\hbar \partial_t \psi = \left(-\frac{\hbar^2}{2m} \partial_x^2 + V \right) \psi.}$$

The path integral is thus a reformulation of the same dynamics, not a new theory.

3.7 Gauge transformations and the phase of the wave function

Source continuation PDF pp. 65–69; SI conventions.

3.7.1 Constant potential versus observable relative phase

If $H' = H + V_0$ everywhere, then $|\psi'(t)\rangle = e^{-iV_0(t-t_0)/\hbar} |\psi(t)\rangle$, a global phase. But two coherent arms exposed to V_I, V_{II} for time T acquire

$$\boxed{\Delta\phi = -\frac{(V_I - V_{II})T}{\hbar}},$$

and the interference contains $1 + \cos \Delta\phi$. The energy zero is arbitrary only when shifted identically for all coherent alternatives; force-free does not mean phase-free.

3.7.2 Electromagnetic gauge covariance

$$\begin{aligned}\mathbf{B} &= \nabla \times \mathbf{A}, & \mathbf{E} &= -\nabla\phi - \partial_t \mathbf{A}, \\ \mathbf{A}' &= \mathbf{A} + \nabla\Lambda, & \phi' &= \phi - \partial_t \Lambda, \\ L &= \frac{1}{2}m\dot{\mathbf{r}}^2 + q\dot{\mathbf{r}} \cdot \mathbf{A} - q\phi, & \mathbf{p}_{\text{can}} &= m\dot{\mathbf{r}} + q\mathbf{A}, & \boldsymbol{\Pi} &= \mathbf{p} - q\mathbf{A}, \\ H &= \frac{(\mathbf{p} - q\mathbf{A})^2}{2m} + q\phi, & [\Pi_i, \Pi_j] &= i\hbar q \epsilon_{ijk} B_k.\end{aligned}$$

Retain the operator order in $(\mathbf{p} - q\mathbf{A})^2$ because $\mathbf{p}$ does not generally commute with $\mathbf{A}(\mathbf{r}, t)$. Gauge covariance requires

$$\boxed{\psi' = e^{iq\Lambda/\hbar}\psi}, \quad (\mathbf{p} - q\mathbf{A}')\psi' = e^{iq\Lambda/\hbar}(\mathbf{p} - q\mathbf{A})\psi.$$

Hence

$$\mathbf{j} = \frac{1}{m} \text{Re}(\psi^* \boldsymbol{\Pi} \psi) = \frac{\hbar}{m} \text{Im}(\psi^* \nabla \psi) - \frac{q}{m} \mathbf{A} |\psi|^2 = \frac{\rho}{m} (\nabla S - q\mathbf{A}).$$

The diamagnetic term is essential; neither ∇S nor $\mathbf{A}$ alone is gauge invariant. For $\mathbf{B} = B\hat{\mathbf{z}}$, the symmetric gauge $\mathbf{A} = \frac{B}{2}(-y, x, 0)$ and Landau gauge $\mathbf{A} = (-By, 0, 0)$ differ by $\nabla(Bxy/2)$ and predict identical observables.

3.7.3 Aharonov–Bohm phase

Two coherent paths can lie where $\mathbf{B} = 0$ yet enclose flux. The action term $q \int_C \mathbf{A} \cdot d\boldsymbol{\ell}$ gives the oriented relative phase

$$\boxed{\Delta\phi = \frac{q}{\hbar} \oint \mathbf{A} \cdot d\boldsymbol{\ell} = \frac{q}{\hbar} \int \mathbf{B} \cdot d\mathbf{S} = \frac{q\Phi_B}{\hbar}}.$$

Only the loop phase/flux, not $\mathbf{A}$ at one point, is gauge invariant. Interference periodicity gives

$$\boxed{\Phi_0 = \frac{h}{|q|}}; \quad \frac{h}{e} \text{ for one electron, } \frac{h}{2e} \text{ for a charge-}2e \text{ condensate.}$$

The effect requires coherence and enclosed flux; it is not a local Lorentz-force effect along either field-free arm.

3.7.4 Magnetic monopole and Dirac quantization

Assume a monopole with the source convention $\Phi_M = \oint \mathbf{B} \cdot d\mathbf{S} = 4\pi g$ and $\mathbf{B} = g\hat{\mathbf{r}}/r^2$. Nonzero flux forbids one global regular vector potential. Use north/south patches:

$$\mathbf{A}_N = g \frac{1 - \cos\theta}{r \sin\theta} \hat{\boldsymbol{\varphi}}, \quad \mathbf{A}_S = -g \frac{1 + \cos\theta}{r \sin\theta} \hat{\boldsymbol{\varphi}}, \quad \mathbf{A}_N - \mathbf{A}_S = \nabla(2g\varphi).$$

On the overlap, $\psi_N = e^{i2gq\varphi/\hbar} \psi_S$. Single-valuedness under $\varphi \mapsto \varphi + 2\pi$ requires

$$e^{i4\pi qg/\hbar} = 1 \implies \boxed{qg = \frac{n\hbar}{2}, \quad n \in \mathbb{Z}} \iff \boxed{q\Phi_M = n\hbar}.$$

The numerical form depends on the definition of g ; retain $\Phi_M = 4\pi g$. This is a conditional consistency result, not evidence that monopoles exist.

3.8 Examples of one-particle dynamics

Source continuation PDF pp. 69–73.

3.8.1 One-dimensional square wells

Infinite centered well: $V = 0$ for $|x| < a$ and $V = \infty$ outside. With $u(\pm a) = 0$ and $k = \sqrt{2mE}/\hbar$,

$$\begin{aligned}u_+(x) &\propto \cos kx, & ka &= (r + \tfrac{1}{2})\pi, & r &= 0, 1, \dots, \\ u_-(x) &\propto \sin kx, & ka &= r\pi, & r &= 1, 2, \dots,\end{aligned}$$

which interleave into

$$E_n = \frac{n^2 \pi^2 \hbar^2}{8ma^2}, \quad n = 1, 2, \dots$$

The width is $2a$; treating a as the width causes a factor-of-four error.

Finite centered well: $V = 0$ for $|x| < a$ and $V = V_0 > 0$ outside. For $0 < E < V_0$ define

$$k = \frac{\sqrt{2mE}}{\hbar}, \quad \kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}, \quad k^2 + \kappa^2 = \frac{2mV_0}{\hbar^2}.$$

Continuity of u, u' at $x = a$ yields

$$k \tan ka = \kappa \quad (\text{even}), \quad -k \cot ka = \kappa \quad (\text{odd}).$$

There are finitely many bound states. Their $e^{-\kappa|x|}$ tails lengthen as $E \rightarrow V_0^-$. For $E > V_0$ the exterior is oscillatory and belongs to the scattering continuum; $V_0 \rightarrow \infty$ recovers the infinite well. If $V(x) = V(-x)$ then $[H, P] = 0$, $P^2 = 1$, and bound states may be chosen even or odd. Under the usual one-dimensional regularity assumptions, bound levels are nondegenerate; alternation additionally follows from node ordering.

3.8.2 Uniform magnetic field: Landau levels

Let a spinless charge q move in the xy plane with $\mathbf{B} = B\hat{\mathbf{z}}$, $B > 0$, and no scalar potential: $H = \mathbf{\Pi}^2/(2m)$, $\omega_c = |q|B/m$. In Landau gauge $\mathbf{A} = (-By, 0, 0)$, $[p_x, H] = 0$. For $\psi = e^{ikx}\phi(y)$,

$$H\phi = \left[\frac{p_y^2}{2m} + \frac{1}{2}m\omega_c^2(y - y_0)^2 \right] \phi, \quad y_0 = -\frac{\hbar k}{qB},$$

so

$$E_n = \hbar\omega_c(n + \tfrac{1}{2}), \quad N_{\text{deg}} = \frac{A|q|B}{h} = \frac{\Phi_B}{h/|q|}$$

(up to edge corrections). The conserved k , equivalently the guiding center y_0 , labels the macroscopic degeneracy. In symmetric gauge let $s = \text{sgn}(qB)$ and

$$a = \frac{\Pi_x + is\Pi_y}{\sqrt{2\hbar|q|B}}, \quad a^\dagger = \frac{\Pi_x - is\Pi_y}{\sqrt{2\hbar|q|B}}.$$

Then $[a, a^\dagger] = 1$ and $H = \hbar\omega_c(a^\dagger a + \frac{1}{2})$. The factor s is essential for a negative charge. Different gauges change labels and phases, not the spectrum or degeneracy. The displayed result neglects spin/Zeman energy, interactions, disorder, confinement, and edge dispersion; in three dimensions add $p_z^2/(2m)$.

3.9 Chapter synthesis

Source continuation PDF pp. 73–74.

- Amplitude and phase obey continuity plus an exact quantum Hamilton–Jacobi equation; Q identifies where classical reasoning fails.
- $K = \langle x|U|x' \rangle$ evolves every state. Composition and time slicing yield $\int \mathcal{D}x e^{iS/\hbar}$; stationary-phase cancellation recovers classical action without discarding quantum paths.
- Gauge transformations change potentials and wave-function phase together, leaving mechanical momentum, current, loop phase, and flux invariant. Aharonov–Bohm interference and Dirac quantization reveal global phase structure.
- Wells and Landau levels exemplify the reusable chain: exploit symmetry or gauge choice, reduce the equation, impose boundary/normalizability conditions, and then read off spectrum and degeneracy.

Self-test for 3.4.1–3.9

1. For $V = k|x|$, derive x_0 and the even/odd Airy conditions. Why is the printed Eq. (3.153) inconsistent?
2. For $V = V_R + iV_I$, which sign of V_I absorbs probability?
3. From $\psi = \sqrt{\rho}e^{iS/\hbar}$ identify $\mathbf{j}$ and Q . Name two places where neglecting Q can fail.
4. State the propagator's initial, composition, and evolution equations. What part of K_0 is not fixed by $e^{iS_{cl}/\hbar}$?
5. Why does the $\xi^2 \partial_x^2 \psi$ term survive the short-time expansion?

6. How do $(\mathbf{A}, \phi, \psi)$ transform, and which phase-gradient combination is gauge invariant?
7. Give the Aharonov–Bohm phase and flux period. Derive Dirac quantization from the north–south transition function.
8. Write the two finite-well bound-state conditions and the Landau energy and degeneracy per area.

Compact answers. **1.** Balance $\hbar^2/(2mx_0^2) = kx_0$; $\text{Ai}'(-\epsilon) = 0$ (even), $\text{Ai}(-\epsilon) = 0$ (odd); the printed factors of two conflict with this x_0 and ϵ . **2.** $V_I < 0$, since the source is $2V_I\rho/\hbar$. **3.** $\mathbf{j} = \rho\nabla S/m$ and $Q = -(\hbar^2/2m)\nabla^2\sqrt{\rho}/\sqrt{\rho}$; e.g. nodes and turning points. **4.** $K \rightarrow \delta$, convolution, and $i\hbar\partial_t K = H_x K$; the normalization and branch/Maslov phase are additional. **5.** Oscillatory-Gaussian scaling gives $\langle \xi^2 \rangle = i\hbar\varepsilon/m = O(\varepsilon)$. **6.** $\mathbf{A}' = \mathbf{A} + \nabla\Lambda$, $\phi' = \phi - \partial_t\Lambda$, $\psi' = e^{iq\Lambda/\hbar}\psi$; $\nabla S - q\mathbf{A}$ is invariant. **7.** $\Delta\phi = q\Phi_B/\hbar$, $\Phi_0 = h/|q|$; single-valued $e^{i2qg\varphi/\hbar}$ gives $qg = n\hbar/2$ or $q\Phi_M = nh$. **8.** $k \tan ka = \kappa$, $-k \cot ka = \kappa$; $E_n = \hbar|q|B(n + 1/2)/m$ and $N_{\text{deg}}/A = |q|B/h$.